

rph8-g15q — formula report

489 inline formulas, 81 display equations, 0 tables, 12 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
rph8-g15q_ E00018 (18)	005	0.837	$\begin{aligned} &\text{\texttt{\textbackslash begin{aligned} \text{ SWAP } L_{1} \text{ SWAP } }} \\ &\text{\texttt{\textbackslash \&=L_{2} \textbackslash\textbackslash \left\{X \otimes X, L_{1}\right\} \textbackslash right\textbackslash} \&=0} \\ &\text{\texttt{\textbackslash end{aligned}}} \end{aligned}$	$\begin{aligned} \text{SWAP } L_1 \text{ SWAP } &= L_2 \\ \{X \otimes X, L_1\} &= 0 \end{aligned}$	$\begin{aligned} \text{SWAP } L_1 \text{ SWAP } &= L_2, \\ \{X \otimes X, L_1\} &= 0 \end{aligned}$
rph8-g15q_ E00065 (D1)	019	0.848	$\begin{aligned} &d\left(x_2, x_1\right) \&=d\ y\left(f\right. \\ &\left.\$		

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rph8-g15q_E00008 (8)	005	■ 0.962	\begin{array}{ll} V_{\{0,0\}}=\mathbb{1}_2 \otimes \mathbb{1}_2 \\ \mathbb{1}_2 \otimes V_{\{1,0\}}=\operatorname{SWAP}, \\ V_{\{0,1\}}=X \otimes X, \text{ \& } V_{\{1,1\}}=\operatorname{SWAP}(X \otimes X) \end{array}	$\begin{array}{ll} V_{(0,0)} = \mathbb{1}_2 \otimes \mathbb{1}_2, & V_{(1,0)} = \text{SWAP}, \\ V_{(0,1)} = X \otimes X, & V_{(1,1)} = \text{SWAP}(X \otimes X), \end{array}$	$\begin{array}{ll} V_{(0,0)} = \mathbb{1}_2 \otimes \mathbb{1}_2, & V_{(1,0)} = \text{SWAP}, \\ V_{(0,1)} = X \otimes X, & V_{(1,1)} = \text{SWAP}(X \otimes X), \end{array}$
rph8-g15q_E00046 (A13)	017	■ 0.985	\begin{aligned} y \text{ \& } &= \left. \frac{d}{dt} \exp(ty) \right _{t=0} = \left. \frac{d}{dt} (f \circ \gamma) \right _{t=0} \text{ by Eq. (A12) } \\ &= Df _{1_G} \left(\left. \frac{d}{dt} \gamma \right _{t=0} \right) \text{ by the chain rule } \\ &= Df _{1_G} \left(\left. \frac{d}{dt} \exp(tx) \right _{t=0} \right) \text{ by Eq. (A11) } \\ &= \mathcal{M}(x) \text{ by definition of } \mathcal{M}. \end{aligned}	$\begin{array}{ll} y = \frac{d}{dt} \exp(ty) \Big _{t=0} = \frac{d}{dt} (f \circ \gamma) \Big _{t=0} & \text{by Eq. (A12)} \\ = Df _{1_G} \left(\left. \frac{d}{dt} \gamma \right _{t=0} \right) & \text{by the chain rule} \\ = Df _{1_G} \left(\left. \frac{d}{dt} \exp(tx) \right _{t=0} \right) & \text{by Eq. (A11)} \\ = \mathcal{M}(x) & \text{by definition of } \mathcal{M} \end{array}$	$\begin{array}{ll} y = \frac{d}{dt} \exp(ty) _{t=0} = \frac{d}{dt} (f \circ \gamma) _{t=0} & \text{by Eq. (A12)} \\ = Df _{1_G} \left(\left. \frac{d}{dt} \gamma _{t=0} \right) \right) & \text{by the chain rule} \\ = Df _{1_G} \left(\left. \frac{d}{dt} \exp(tx) _{t=0} \right) \right) & \text{by Eq. (A11)} \\ = \mathcal{M}(x) & \text{by definition of } \mathcal{M}. \end{array} \quad (A)$
rph8-g15q_E00067 (D3)	019	■ 0.991	\begin{aligned} d_{\{\mathrm{X}\}} \left(x_1, x_2 \right) &\geq d_Y \left(f \left(x_1 \right), f \left(x_3 \right) \right) + d_Y \left(f \left(x_3 \right), f \left(x_2 \right) \right) \\ &= d_{\mathcal{X}} \left(x_1, x_3 \right) + d_{\mathcal{X}} \left(x_3, x_2 \right), \end{aligned}	$\begin{array}{l} d_{\mathcal{X}} \left(x_1, x_2 \right) \geq d_Y \left(f \left(x_1 \right), f \left(x_3 \right) \right) + d_Y \left(f \left(x_3 \right), f \left(x_2 \right) \right) \\ = d_{\mathcal{X}} \left(x_1, x_3 \right) + d_{\mathcal{X}} \left(x_3, x_2 \right) \end{array}$	$\begin{array}{l} d_{\mathcal{X}}(x_1, x_2) \geq d_Y(f(x_1), f(x_3)) + d_Y(f(x_3), f(x_2)) \\ = d_{\mathcal{X}}(x_1, x_3) + d_{\mathcal{X}}(x_3, x_2), \end{array}$
rph8-g15q_E00027 (27)	010	■ 0.994	d_{\mathrm{Tr}}(\rho, \sigma) = \ \rho - \sigma\ _1 \equiv \mathrm{Tr} \left[\sqrt{(\rho - \sigma)^\dagger (\rho - \sigma)} \right].	$d_{\mathrm{Tr}}(\rho, \sigma) = \ \rho - \sigma\ _1 \equiv \mathrm{Tr} \left[\sqrt{(\rho - \sigma)^\dagger (\rho - \sigma)} \right].$	$d_{\mathrm{Tr}}(\rho, \sigma) = \ \rho - \sigma\ _1 \equiv \mathrm{Tr} \left[\sqrt{(\rho - \sigma)^\dagger (\rho - \sigma)} \right].$
rph8-g15q_E00054 (B5)	018	■ 0.994	\left \psi_0 \right\rangle = \sqrt{p} +, +\rangle - \sqrt{1-p} -, -\rangle	$ \psi_0\rangle = \sqrt{p} +,+\rangle - \sqrt{1-p} -, -\rangle$	$ \psi_0\rangle = \sqrt{p} +,+\rangle - \sqrt{1-p} -, -\rangle$
rph8-g15q_E00009 (9)	005	■ 0.996	U \left(x_1, x_2 \right) = R_Z \left(x_1 \right) \otimes R_Z \left(x_2 \right),	$U \left(x_1, x_2 \right) = R_Z \left(x_1 \right) \otimes R_Z \left(x_2 \right),$	$U(x_1, x_2) = R_Z(x_1) \otimes R_Z(x_2),$
rph8-g15q_E00011 (11)	005	■ 0.999	\left \psi_0 \right\rangle = \sqrt{p} +, +\rangle - \sqrt{1-p} -, -\rangle,	$ \psi_0\rangle = \sqrt{p} +,+\rangle - \sqrt{1-p} -, -\rangle,$	$ \psi_0\rangle = \sqrt{p} +,+\rangle - \sqrt{1-p} -, -\rangle,$
rph8-g15q_E00005 (5)	005	■ 1.000	y \left(x_1, x_2 \right) = y \left(x_2, x_1 \right) = y \left(-x_1, -x_2 \right),	$y \left(x_1, x_2 \right) = y \left(x_2, x_1 \right) = y \left(-x_1, -x_2 \right),$	$y(x_1, x_2) = y(x_2, x_1) = y(-x_1, -x_2),$
rph8-g15q_E00016 (16)	005	■ 1.000	V_g e^{-i \mathcal{L}(x)} V_g^\dagger = e^{-i \mathcal{L}(\alpha_g(x))}	$V_g e^{-i \mathcal{L}(x)} V_g^\dagger = e^{-i \mathcal{L}(\alpha_g(x))}$	$V_g e^{-i \mathcal{L}(x)} V_g^\dagger = e^{-i \mathcal{L}(\alpha_g(x))}$
rph8-g15q_E00023 (23)	007	■ 1.000	x \mapsto \chi(x) = \sum_{n=0}^{2^d-1} x_n n\rangle \mapsto \Phi(x)\rangle = \frac{\chi(x)}{\ \chi(x)\ },	$x \mapsto \chi(x) = \sum_{n=0}^{2^d-1} x_n n\rangle \mapsto \Phi(x)\rangle = \frac{\chi(x)}{\ \chi(x)\ },$	$x \mapsto \chi(x) = \sum_{n=0}^{2^d-1} x_n n\rangle \mapsto \Phi(x)\rangle = \frac{\chi(x)}{\ \chi(x)\ },$
rph8-g15q_E00029 (29)	010	■ 1.000	d_B(\rho, \sigma) = \sqrt{2(1 - \sqrt{F(\rho, \sigma)})},	$d_B(\rho, \sigma) = \sqrt{2(1 - \sqrt{F(\rho, \sigma)})},$	$d_B(\rho, \sigma) = \sqrt{2(1 - \sqrt{F(\rho, \sigma)})},$
rph8-g15q_E00030 (30)	010	■ 1.000	F(\rho, \sigma) = (\mathrm{Tr}[\sqrt{\rho\sigma\sqrt{\rho}}])^2	$F(\rho, \sigma) = (\mathrm{Tr}[\sqrt{\rho\sigma\sqrt{\rho}}])^2$	$F(\rho, \sigma) = \left(\mathrm{Tr} \left[\sqrt{\rho\sigma\sqrt{\rho}} \right] \right)^2$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					

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rph8-g15q_E00032 (32)	013	■ 1.000	$F\left(\operatorname{id}_{\left(\operatorname{\mathcal{X}},\operatorname{\mathcal{A}}\right)}\right)=\operatorname{id}_{\left(\operatorname{\mathcal{X}},\operatorname{\mathcal{A}}\right)}=\operatorname{id}_{\operatorname{\mathcal{X}}}=\operatorname{id}_{F\left(\operatorname{\mathcal{X}},\operatorname{\mathcal{A}}\right)}.$	$F\left(\operatorname{id}_{\left(\mathcal{X},\alpha\right)}\right)=\operatorname{id}_{\left(\mathcal{X},\alpha\right)}=\operatorname{id}_{\mathcal{X}}=\operatorname{id}_{F\left(\mathcal{X},\alpha\right)}.$	$F(\operatorname{id}_{(\mathcal{X},\alpha)})=\operatorname{id}_{(\mathcal{X},\alpha)}=\operatorname{id}_{\mathcal{X}}=\operatorname{id}_{F(\mathcal{X},\alpha)}.$
rph8-g15q_E00001 (1)	003	■ 1.000	$\rho(x)=U(x)\left \psi_0\right\rangle\left\langle\psi_0\right U(x)^{\dagger}.$	$\rho(x)=U(x)\left \psi_0\right\rangle\left\langle\psi_0\right U(x)^{\dagger}.$	$\rho(x)=U(x) \psi_0\rangle\langle\psi_0 U(x)^{\dagger}.$
rph8-g15q_E00002 (2)	004	■ 1.000	$\operatorname{Ad}_{V_g}(\sigma):=V_g\sigma V_g^{\dagger}$	$\operatorname{Ad}_{V_g}(\sigma):=V_g\sigma V_g^{\dagger}$	$\operatorname{Ad}_{V_g}(\sigma):=V_g\sigma V_g^{\dagger}$
rph8-g15q_E00003 (3)	004	■ 1.000	$\rho\left(\alpha_g(x)\right)=V_g\rho(x)V_g^{\dagger}$	$\rho\left(\alpha_g(x)\right)=V_g\rho(x)V_g^{\dagger}$	$\rho(\alpha_g(x))=V_g\rho(x)V_g^{\dagger}$
rph8-g15q_E00004 (4)	004	■ 1.000	$f\left(\alpha_g(x)\right)=\beta_g(f(x))$	$f\left(\alpha_g(x)\right)=\beta_g(f(x))$	$f(\alpha_g(x))=\beta_g(f(x))$
rph8-g15q_E00006 (6)	005	■ 1.000	$G=\mathbb{Z}_2\times\mathbb{Z}_2=\left\{\left(0,0\right),\left(1,0\right),\left(0,1\right),\left(1,1\right)\right\}$	$G=\mathbb{Z}_2\times\mathbb{Z}_2=\left\{\left(0,0\right),\left(1,0\right),\left(0,1\right),\left(1,1\right)\right\}$	$G=\mathbb{Z}_2\times\mathbb{Z}_2=\left\{\left(0,0\right),\left(1,0\right),\left(0,1\right),\left(1,1\right)\right\}$
rph8-g15q_E00010 (10)	005	■ 1.000	$R_Z(\theta)=e^{-i\theta Z/2}=\begin{bmatrix}e^{-i\theta/2}&0\\0&e^{i\theta/2}\end{bmatrix}$	$R_Z(\theta)=e^{-i\theta Z/2}=\begin{bmatrix}e^{-i\theta/2}&0\\0&e^{i\theta/2}\end{bmatrix}$	$R_Z(\theta)=e^{-i\theta Z/2}=\begin{bmatrix}e^{-i\theta/2}&0\\0&e^{i\theta/2}\end{bmatrix},$
rph8-g15q_E00012 (12)	005	■ 1.000	$\left +\right\rangle=\frac{1}{\sqrt{2}}\left(\left 0\right\rangle+\left 1\right\rangle\right)\quad\text{and}\quad\left -\right\rangle=\frac{1}{\sqrt{2}}\left(\left 0\right\rangle-\left 1\right\rangle\right)$	$\left +\right\rangle=\frac{1}{\sqrt{2}}(0\rangle+ 1\rangle)\quad\text{and}\quad\left -\right\rangle=\frac{1}{\sqrt{2}}(0\rangle- 1\rangle)$	$\left +\right\rangle=\frac{1}{\sqrt{2}}(0\rangle+ 1\rangle)\quad\text{and}\quad\left -\right\rangle=\frac{1}{\sqrt{2}}(0\rangle- 1\rangle)$
rph8-g15q_E00013 (13)	005	■ 1.000	$\rho(x)=U\left(x_1,x_2\right)\left \psi_0\right\rangle\left\langle\psi_0\right U\left(x_1,x_2\right)^{\dagger}.$	$\rho(x)=U\left(x_1,x_2\right) \psi_0\rangle\langle\psi_0 U\left(x_1,x_2\right)^{\dagger}.$	$\rho(x)=U(x_1,x_2) \psi_0\rangle\langle\psi_0 U(x_1,x_2)^{\dagger}.$
rph8-g15q_E00014 (14)	005	■ 1.000	$U(x)=e^{-i\operatorname{\mathcal{L}}(x)},$	$U(x)=e^{-i\mathcal{L}(x)},$	$U(x)=e^{-i\mathcal{L}(x)},$
rph8-g15q_E00015 (15)	005	■ 1.000	$L_1:=\operatorname{\mathcal{L}}(e_1)\quad\text{and}\quad L_2:=\operatorname{\mathcal{L}}(e_2),$	$L_1:=\mathcal{L}(e_1)\quad\text{and}\quad L_2:=\mathcal{L}(e_2),$	$L_1:=\mathcal{L}(e_1)\quad\text{and}\quad L_2:=\mathcal{L}(e_2),$
rph8-g15q_E00017 (17)	005	■ 1.000	$V_g\operatorname{\mathcal{L}}(x)V_g^{\dagger}=\operatorname{\mathcal{L}}(\alpha_g(x))$	$V_g\mathcal{L}(x)V_g^{\dagger}=\mathcal{L}(\alpha_g(x))$	$V_g\mathcal{L}(x)V_g^{\dagger}=\mathcal{L}(\alpha_g(x))$
rph8-g15q_E00020 (20)	006	■ 1.000	$L_1=\frac{Z}{2}\otimes\mathbb{1}_2\quad\text{and}\quad L_2=\mathbb{1}_2\otimes\frac{Z}{2}$	$L_1=\frac{Z}{2}\otimes\mathbb{1}_2\quad\text{and}\quad L_2=\mathbb{1}_2\otimes\frac{Z}{2}$	$L_1=\frac{Z}{2}\otimes\mathbb{1}_2\quad\text{and}\quad L_2=\mathbb{1}_2\otimes\frac{Z}{2}$
rph8-g15q_E00021 (21)	006	■ 1.000	$x\mapsto\left \Phi(x)\right\rangle:=\left(\bigotimes_{k=1}^d\exp\left(-\frac{ix_k}{2}X_k\right)\right)\left 0\cdots0\right\rangle,$	$x\mapsto \Phi(x)\rangle:=\left(\bigotimes_{k=1}^d\exp\left(-\frac{ix_k}{2}X_k\right)\right) 0\cdots0\rangle,$	$x\mapsto \Phi(x)\rangle:=\left(\bigotimes_{k=1}^d\exp\left(-\frac{ix_k}{2}X_k\right)\right) 0\cdots0\rangle,$
rph8-g15q_E00022 (22)	006	■ 1.000	$\rho(x):=\left \Phi(x)\right\rangle\left\langle\Phi(x)\right .$	$\rho(x):= \Phi(x)\rangle\langle\Phi(x) .$	$\rho(x):= \Phi(x)\rangle\langle\Phi(x) .$
rph8-g15q_E00024 (24)	007	■ 1.000	$(x_0,x_1,\dots,x_{N-1})\mapsto\frac{(x_0,x_1,\dots,x_{N-1},1)}{\sqrt{1+\ x\ ^2}},$	$(x_0,x_1,\dots,x_{N-1})\mapsto\frac{(x_0,x_1,\dots,x_{N-1},1)}{\sqrt{1+\ x\ ^2}},$	$(x_0,x_1,\dots,x_{N-1})\mapsto\frac{(x_0,x_1,\dots,x_{N-1},1)}{\sqrt{1+\ x\ ^2}},$
rph8-g15q_E00025 (25)	007	■ 1.000	$(x_0,\dots,x_{N-1})\mapsto\frac{(e^{x_0},\dots,e^{x_{N-1}},1)}{1+\sum_{j=0}^{N-1}e^{x_j}},$	$(x_0,\dots,x_{N-1})\mapsto\frac{(e^{x_0},\dots,e^{x_{N-1}},1)}{1+\sum_{j=0}^{N-1}e^{x_j}},$	$(x_0,\dots,x_{N-1})\mapsto\frac{(e^{x_0},\dots,e^{x_{N-1}},1)}{1+\sum_{j=0}^{N-1}e^{x_j}},$
rph8-g15q_E00026 (26)	009	■ 1.000	$d_{\operatorname{\mathcal{X}}}\left(x_1,x_2\right):=d_{\operatorname{\mathcal{Y}}}\left(f\left(x_1\right),f\left(x_2\right)\right)$	$d_{\mathcal{X}}\left(x_1,x_2\right):=d_{\mathcal{Y}}\left(f\left(x_1\right),f\left(x_2\right)\right)$	$d_{\mathcal{X}}(x_1,x_2):=d_{\mathcal{Y}}(f(x_1),f(x_2))$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value					

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rph8-g15q_ EQ0028 (28)	010	■ 1.000	$d_{\mathrm{HS}}(\rho, \sigma) = \sqrt{\mathrm{Tr}[(\rho - \sigma)^\dagger(\rho - \sigma)]}.$	$d_{\mathrm{HS}}(\rho, \sigma) = \sqrt{\mathrm{Tr}[(\rho - \sigma)^\dagger(\rho - \sigma)]}.$	$d_{\mathrm{HS}}(\rho, \sigma) = \sqrt{\mathrm{Tr}[(\rho - \sigma)^\dagger(\rho - \sigma)]}.$
rph8-g15q_ EQ0031 (31)	013	■ 1.000	$F(g \circ f) = g \circ f = F(g) \circ F(f).$	$F(g \circ f) = g \circ f = F(g) \circ F(f).$	$F(g \circ f) = g \circ f = F(g) \circ F(f).$
rph8-g15q_ EQ0033 (33)	014	■ 1.000	$U(x) = e^{-i \mathcal{L}(x)},$	$U(x) = e^{-i \mathcal{L}(x)},$	$U(x) = e^{-i \mathcal{L}(x)},$
rph8-g15q_ EQ0034 (A1)	016	■ 1.000	$U(x) = e^{-i \mathcal{L}(x)},$	$U(x) = e^{-i \mathcal{L}(x)},$	$U(x) = e^{-i \mathcal{L}(x)},$
rph8-g15q_ EQ0035 (A2)	016	■ 1.000	$\gamma(t) = \exp(tM)$	$\gamma(t) = \exp(tM)$	$\gamma(t) = \exp(tM)$
rph8-g15q_ EQ0036 (A3)	016	■ 1.000	$f(\exp(W)) = \exp(\mathcal{M}(W))$	$f(\exp(W)) = \exp(\mathcal{M}(W))$	$f(\exp(W)) = \exp(\mathcal{M}(W))$
rph8-g15q_ EQ0037 (A4)	016	■ 1.000	$\left[\mathcal{M}(W_1), \mathcal{M}(W_2)\right] = \mathcal{M}([W_1, W_2])$	$[\mathcal{M}(W_1), \mathcal{M}(W_2)] = \mathcal{M}([W_1, W_2])$	$[\mathcal{M}(W_1), \mathcal{M}(W_2)] = \mathcal{M}([W_1, W_2])$
rph8-g15q_ EQ0038 (A5)	017	■ 1.000	$U((s, t)) = e^{-isX - itY}$	$U((s, t)) = e^{-isX - itY}$	$U((s, t)) = e^{-isX - itY}$
rph8-g15q_ EQ0039 (A6)	017	■ 1.000	$\begin{aligned} U((s, 0))U((0, t)) &= e^{-isX}e^{-itY} \\ &\neq e^{-i(sX + tY)} \\ &= U((s, 0) + (0, t)) \end{aligned}$	$\begin{aligned} U((s, 0))U((0, t)) &= e^{-isX}e^{-itY} \\ &\neq e^{-i(sX + tY)} \\ &= U((s, 0) + (0, t)) \end{aligned}$	$\begin{aligned} U((s, 0))U((0, t)) &= e^{-isX}e^{-itY} \\ &\neq e^{-i(sX + tY)} \\ &= U((s, 0) + (0, t)) \end{aligned}$
rph8-g15q_ EQ0040 (A7)	017	■ 1.000	$\mathbb{R}_\alpha = \{r\alpha : \alpha \in \mathbb{R}^2 \setminus \{0\}, r \in \mathbb{R}\},$	$\mathbb{R}_\alpha = \{r\alpha : \alpha \in \mathbb{R}^2 \setminus \{0\}, r \in \mathbb{R}\},$	$\mathbb{R}_\alpha = \{r\alpha : \alpha \in \mathbb{R}^2 \setminus \{0\}, r \in \mathbb{R}\},$
rph8-g15q_ EQ0041 (A8)	017	■ 1.000	$\begin{aligned} U(r\alpha)U(u\alpha) &= e^{-irW}e^{-iuW} \\ &= e^{-i(r+u)W} \\ &= U(r\alpha + u\alpha), \end{aligned}$	$\begin{aligned} U(r\alpha)U(u\alpha) &= e^{-irW}e^{-iuW} \\ &= e^{-i(r+u)W} \\ &= U(r\alpha + u\alpha), \end{aligned}$	$\begin{aligned} U(r\alpha)U(u\alpha) &= e^{-irW}e^{-iuW} \\ &= e^{-i(r+u)W} \\ &= U(r\alpha + u\alpha), \end{aligned}$
rph8-g15q_ EQ0042 (A9)	017	■ 1.000	$\mathbb{R} \xrightarrow{\alpha} \mathbb{R}^2 \xrightarrow{U} \mathcal{U}(\mathbb{C}^2)$	$\mathbb{R} \xrightarrow{\alpha} \mathbb{R}^2 \xrightarrow{U} \mathcal{U}(\mathbb{C}^2)$	$\mathbb{R} \xrightarrow{\alpha} \mathbb{R}^2 \xrightarrow{U} \mathcal{U}(\mathbb{C}^2)$
rph8-g15q_ EQ0043 (A10)	017	■ 1.000	$f(\exp(x)) = \exp(\mathcal{M}(x))$	$f(\exp(x)) = \exp(\mathcal{M}(x))$	$f(\exp(x)) = \exp(\mathcal{M}(x))$
rph8-g15q_ EQ0044 (A11)	017	■ 1.000	$\gamma(t) = \exp(tx)$	$\gamma(t) = \exp(tx)$	$\gamma(t) = \exp(tx)$
rph8-g15q_ EQ0045 (A12)	017	■ 1.000	$(f \circ \gamma)(t) = \exp(ty)$	$(f \circ \gamma)(t) = \exp(ty)$	$(f \circ \gamma)(t) = \exp(ty)$
rph8-g15q_ EQ0047 (A14)	017	■ 1.000	$f(\exp(tx)) = \exp(t\mathcal{M}(x))$	$f(\exp(tx)) = \exp(t\mathcal{M}(x))$	$f(\exp(tx)) = \exp(t\mathcal{M}(x))$
rph8-g15q_ EQ0048 (A15)	018	■ 1.000	$f(x) = \exp(\mathcal{M}(x))$	$f(x) = \exp(\mathcal{M}(x))$	$f(x) = \exp(\mathcal{M}(x))$
rph8-g15q_ EQ0050 (B1)	018	■ 1.000	$c(x_1, x_2) = c(x_2, x_1) = c(-x_1, -x_2).$	$c(x_1, x_2) = c(x_2, x_1) = c(-x_1, -x_2).$	$c(x_1, x_2) = c(x_2, x_1) = c(-x_1, -x_2).$
rph8-g15q_ EQ0051 (B2)	018	■ 1.000	$c(x) = \begin{cases} -1, & \text{if } y(x) < 0, \\ 0, & \text{if } y(x) = 0, \\ +1, & \text{if } y(x) > 0, \end{cases}$	$c(x) = \begin{cases} -1, & \text{if } y(x) < 0, \\ 0, & \text{if } y(x) = 0, \\ +1, & \text{if } y(x) > 0, \end{cases}$	$c(x) = \begin{cases} -1, & \text{if } y(x) < 0, \\ 0, & \text{if } y(x) = 0, \\ +1, & \text{if } y(x) > 0, \end{cases}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					

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rph8-g15q_ EQ0052 (B3)	018	■ 1.000	$y(x)=\operatorname{Tr}[\rho(x) \, O],$	$y(x) = \operatorname{Tr}[\rho(x)O],$	$y(x) = \operatorname{Tr}[\rho(x)O],$
rph8-g15q_ EQ0053 (B4)	018	■ 1.000	$y\left(x_{\{1\}},\,x_{\{2\}}\right)=y\left(x_{\{2\}},\,x_{\{1\}}\right)=y\left(-x_{\{1\}},-x_{\{2\}}\right)$	$y\left(x_1,x_2\right)=y\left(x_2,x_1\right)=y\left(-x_1,-x_2\right)$	$y(x_1,x_2)=y(x_2,x_1)=y(-x_1,-x_2)$
rph8-g15q_ EQ0055 (B6)	018	■ 1.000	$0=X\,\otimes\,X,$	$O = X \otimes X,$	$O = X \otimes X,$
rph8-g15q_ EQ0056 (B7)	018	■ 1.000	$y\left(x_{\{1\}},\,x_{\{2\}}\right)=\cos\left(x_{\{1\}}\right)\sin\left(x_{\{2\}}\right)+2\sqrt{p\left(1-p\right)}\sin\left(x_{\{1\}}\right)\sin\left(x_{\{2\}}\right).$	$y\left(x_1,x_2\right)=\cos\left(x_1\right)\sin\left(x_2\right)+2\sqrt{p\left(1-p\right)}\sin\left(x_1\right)\sin\left(x_2\right).$	$y(x_1,x_2)=\cos(x_1)\sin(x_2)+2\sqrt{p(1-p)}\sin(x_1)\sin(x_2).$
rph8-g15q_ EQ0057 (B8)	018	■ 1.000	$L_{\{1\}}=\left[\begin{array}{llll} a_{\{11\}} & a_{\{12\}} & a_{\{13\}} & a_{\{14\}} \\ & \overline{a_{\{12\}}} & a_{\{22\}} & a_{\{23\}} \\ & \overline{a_{\{13\}}} & \overline{a_{\{23\}}} & a_{\{33\}} \\ & \overline{a_{\{14\}}} & \overline{a_{\{24\}}} & \overline{a_{\{34\}}} \end{array}\right],$	$L_1 = \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \end{array}\right],$	$L_1 = \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \end{array}\right],$
rph8-g15q_ EQ0058 (C3)	019	■ 1.000	$\left[\begin{array}{llll} \overline{a_{\{14\}}} & \overline{a_{\{24\}}} & \overline{a_{\{34\}}} & a_{44} \\ \overline{a_{\{13\}}} & \overline{a_{\{23\}}} & a_{33} & a_{34} \\ \overline{a_{\{12\}}} & a_{22} & a_{23} & a_{24} \\ a_{11} & a_{12} & a_{13} & a_{14} \end{array}\right] = - \left[\begin{array}{cccc} a_{14} & a_{13} & a_{12} & a_{11} \\ a_{24} & a_{23} & a_{22} & \overline{a_{12}} \\ a_{34} & a_{33} & \overline{a_{23}} & \overline{a_{13}} \\ a_{44} & \overline{a_{34}} & \overline{a_{24}} & \overline{a_{14}} \end{array}\right].$	$\left[\begin{array}{cccc} \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ a_{11} & a_{12} & a_{13} & a_{14} \end{array}\right] = - \left[\begin{array}{cccc} a_{14} & a_{13} & a_{12} & a_{11} \\ a_{24} & a_{23} & a_{22} & \overline{a_{12}} \\ a_{34} & a_{33} & \overline{a_{23}} & \overline{a_{13}} \\ a_{44} & \overline{a_{34}} & \overline{a_{24}} & \overline{a_{14}} \end{array}\right].$	$\left[\begin{array}{cccc} \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ a_{11} & a_{12} & a_{13} & a_{14} \end{array}\right] = - \left[\begin{array}{cccc} a_{14} & a_{13} & a_{12} & a_{11} \\ a_{24} & a_{23} & a_{22} & \overline{a_{12}} \\ a_{34} & a_{33} & \overline{a_{23}} & \overline{a_{13}} \\ a_{44} & \overline{a_{34}} & \overline{a_{24}} & \overline{a_{14}} \end{array}\right].$
rph8-g15q_ EQ0059 (B10)	019	■ 1.000	$L_{\{1\}}=\left[\begin{array}{cccc} a_{\{11\}} & a_{\{12\}} & a_{\{13\}} & a_{\{14\}} \\ & \overline{a_{\{12\}}} & a_{\{22\}} & a_{\{23\}} \\ & \overline{a_{\{13\}}} & \overline{a_{\{23\}}} & -a_{\{23\}} \\ & -a_{\{22\}} & -\overline{a_{\{12\}}} & -a_{\{14\}} \\ & -a_{\{12\}} & -a_{\{11\}} \end{array}\right],$	$L_1 = \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & -\overline{a_{13}} \\ \overline{a_{13}} & -a_{23} & -a_{22} & -\overline{a_{12}} \\ -a_{14} & -a_{13} & -a_{12} & -a_{11} \end{array}\right],$	$L_1 = \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & -\overline{a_{13}} \\ \overline{a_{13}} & -a_{23} & -a_{22} & -\overline{a_{12}} \\ -a_{14} & -a_{13} & -a_{12} & -a_{11} \end{array}\right],$
rph8-g15q_ EQ0060 (B11)	019	■ 1.000	$a_{\{12\}},\,a_{\{13\}}\in\mathbb{C},\,\,\,\text{quad}\,\,a_{\{14\}},\,a_{\{23\}}\in i\mathbb{R},\,\,\,\text{quad}\,\,a_{\{11\}},\,a_{\{22\}}\in\mathbb{R}.$	$a_{12},a_{13}\in\mathbb{C},\quad a_{14},a_{23}\in i\mathbb{R},\quad a_{11},a_{22}\in\mathbb{R}.$	$a_{12},a_{13}\in\mathbb{C},\quad a_{14},a_{23}\in i\mathbb{R},\quad a_{11},a_{22}\in\mathbb{R}.$
rph8-g15q_ EQ0062 (C1)	019	■ 1.000	$d_{\mathcal{X}}\left(x_{\{1\}},\,x_{\{2\}}\right)=\sqrt{\left(x_{\{1\}}-x_{\{2\}}\right)^T\left(V^T V\right)\left(x_{\{1\}}-x_{\{2\}}\right)},$	$d_{\mathcal{X}}\left(x_1,x_2\right)=\sqrt{\left(x_1-x_2\right)^T\left(V^T V\right)\left(x_1-x_2\right)},$	$d_{\mathcal{X}}(x_1,x_2)=\sqrt{(x_1-x_2)^T(V^TV)(x_1-x_2)},$
rph8-g15q_ EQ0063 (C2)	019	■ 1.000	$d_{\{A\}}\left(x_{\{1\}},\,x_{\{2\}}\right):=\sqrt{\left(x_{\{1\}}-x_{\{2\}}\right)^T A\left(x_{\{1\}}-x_{\{2\}}\right)}.$	$d_A\left(x_1,x_2\right):=\sqrt{\left(x_1-x_2\right)^T A\left(x_1-x_2\right)}.$	$d_A(x_1,x_2):=\sqrt{(x_1-x_2)^TA(x_1-x_2)}.$
rph8-g15q_ EQ0066 (D2)	019	■ 1.000	$\begin{array}{l} d_{\mathcal{X}}\left(x_{\{1\}},\,x_{\{2\}}\right) \\ \quad = d_{\mathcal{Y}}\left(f\left(x_{\{1\}}\right),\,f\left(x_{\{2\}}\right)\right) \\ \quad \geq d_{\mathcal{Y}}\left(f\left(x_{\{1\}}\right),\,y\right)+d_{\mathcal{Y}}\left(y,\,f\left(x_{\{2\}}\right)\right) \end{array}$	$\begin{array}{l} d_{\mathcal{X}}\left(x_1,x_2\right)=d_{\mathcal{Y}}\left(f\left(x_1\right),\,f\left(x_2\right)\right) \\ \qquad \qquad \qquad \geq d_{\mathcal{Y}}\left(f\left(x_1\right),\,y\right)+d_{\mathcal{Y}}\left(y,\,f\left(x_2\right)\right) \end{array}$	$\begin{array}{l} d_{\mathcal{X}}(x_1,x_2)=d_{\mathcal{Y}}(f(x_1),f(x_2)) \\ \qquad \qquad \qquad \geq d_{\mathcal{Y}}(f(x_1),y)+d_{\mathcal{Y}}(y,f(x_2)) \end{array}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
rph8-g15q_ EQ0068 (D4)	019	■ 1.000	$\theta=\mathrm{d}_{\mathcal{X}}\left(x_1, x_2\right)=\mathrm{d}_{\mathcal{Y}}\left(f\left(x_1\right), f\left(x_2\right)\right),$	$0 = d_{\mathcal{X}}(x_1, x_2) = d_{\mathcal{Y}}(f(x_1), f(x_2)),$	$0 = d_{\mathcal{X}}(x_1, x_2) = d_{\mathcal{Y}}(f(x_1), f(x_2)),$
rph8-g15q_ EQ0069 (D5)	020	■ 1.000	$c(x)=\begin{cases}+1, & \text{if } x \in A, \\ -1, & \text{if } x \in B.\end{cases}$	$c(x) = \begin{cases} +1, & \text{if } x \in A, \\ -1, & \text{if } x \in B. \end{cases}$	$c(x) = \begin{cases} +1, & \text{if } x \in A, \\ -1, & \text{if } x \in B. \end{cases}$
rph8-g15q_ EQ0070 (D6)	020	■ 1.000	$\rho_{\mathrm{A}}:=\frac{1}{\#\mathrm{A}}\sum_{\mathrm{a}\in\mathrm{A}}\rho(\mathrm{a})\quad\text{and}\quad\rho_{\mathrm{B}}:=\frac{1}{\#\mathrm{B}}\sum_{\mathrm{b}\in\mathrm{B}}\rho(\mathrm{b}),$	$\rho_A := \frac{1}{\#A} \sum_{a \in A} \rho(a) \quad \text{and} \quad \rho_B := \frac{1}{\#B} \sum_{b \in B} \rho(b),$	$\rho_A := \frac{1}{\#A} \sum_{a \in A} \rho(a) \quad \text{and} \quad \rho_B := \frac{1}{\#B} \sum_{b \in B} \rho(b),$
rph8-g15q_ EQ0071 (D7)	020	■ 1.000	$C\left(\rho_{\mathrm{A}}, \rho_{\mathrm{B}}\right)=1-\frac{1}{2}d_{\mathrm{HS}}\left(\rho_{\mathrm{A}}, \rho_{\mathrm{B}}\right).$	$C(\rho_A, \rho_B) = 1 - \frac{1}{2} d_{\mathrm{HS}}(\rho_A, \rho_B).$	$C(\rho_A, \rho_B) = 1 - \frac{1}{2} d_{\mathrm{HS}}(\rho_A, \rho_B).$
rph8-g15q_ EQ0072 (D8)	020	■ 1.000	$U_{\theta}(x)=\left(\prod_{n=1}^4R_X(x)R_Y\left(\theta_n\right)\right)R_X(x),$	$U_{\theta}(x) = \left(\prod_{n=1}^4 R_X(x) R_Y(\theta_n)\right) R_X(x),$	$U_{\theta}(x) = \left(\prod_{n=1}^4 R_X(x) R_Y(\theta_n)\right) R_X(x),$
rph8-g15q_ EQ0073 (D9)	020	■ 1.000	$\prod_{n=1}^N A_n=A_N\cdots A_1.$	$\prod_{n=1}^N A_n = A_N \cdots A_1.$	$\prod_{n=1}^N A_n = A_N \cdots A_1.$
rph8-g15q_ EQ0074 (D10)	020	■ 1.000	$\rho_{\theta}(x)= x\rangle\langle x ,$	$\rho_{\theta}(x) = x\rangle\langle x ,$	$\rho_{\theta}(x) = x\rangle\langle x ,$
rph8-g15q_ EQ0075 (D11)	020	■ 1.000	$ x\rangle=U_{\theta}(x) 0\rangle.$	$ x\rangle = U_{\theta}(x) 0\rangle.$	$ x\rangle = U_{\theta}(x) 0\rangle.$
rph8-g15q_ EQ0076 (D12)	020	■ 1.000	$O:=\rho_{\mathrm{A}}-\rho_{\mathrm{B}},$	$O := \rho_A - \rho_B,$	$O := \rho_A - \rho_B,$
rph8-g15q_ EQ0077 (D13)	020	■ 1.000	$y(x)=\operatorname{Tr}[\rho(x)O].$	$y(x) = \operatorname{Tr}[\rho(x)O].$	$y(x) = \operatorname{Tr}[\rho(x)O].$
rph8-g15q_ EQ0078 (D14)	020	■ 1.000	$\operatorname{Tr}\left[\rho_{\mathrm{A}}-\rho_{\mathrm{B}}\right]=\operatorname{Tr}\left[\rho_{\mathrm{A}}\right]-\operatorname{Tr}\left[\rho_{\mathrm{B}}\right]=1-1=0,$	$\operatorname{Tr}[\rho_A - \rho_B] = \operatorname{Tr}[\rho_A] - \operatorname{Tr}[\rho_B] = 1 - 1 = 0,$	$\operatorname{Tr}[\rho_A - \rho_B] = \operatorname{Tr}[\rho_A] - \operatorname{Tr}[\rho_B] = 1 - 1 = 0,$
rph8-g15q_ EQ0079 (D15)	020	■ 1.000	$\operatorname{sgn}(y(x))=c(x)$	$\operatorname{sgn}(y(x)) = c(x)$	$\operatorname{sgn}(y(x)) = c(x)$
rph8-g15q_ EQ0080 (E2)	021	■ 1.000	$\overbrace{\mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2}^{n \text{ times}} \cong \mathbb{C}^{2^n},$	$\overbrace{\mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2}^{n \text{ times}} \cong \mathbb{C}^{2^n},$	$\overbrace{\mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2}^{n \text{ times}} \cong \mathbb{C}^{2^n},$
rph8-g15q_ EQ0081 (E3)	021	■ 1.000	$\left x_{\mathrm{0}}\, x_{\mathrm{1}}\, \cdots\, x_{\mathrm{n-1}}\right\rangle\!\rangle=\left x_{\mathrm{0}}\right\rangle\!\rangle\otimes\left x_{\mathrm{1}}\right\rangle\!\rangle\cdots\otimes\left x_{\mathrm{n-1}}\right\rangle\!\rangle.$	$ x_0 x_1 \cdots x_{n-1}\rangle = x_0\rangle \otimes x_1\rangle \otimes \cdots \otimes x_{n-1}\rangle.$	$ x_0 x_1 \cdots x_{n-1}\rangle = x_0\rangle \otimes x_1\rangle \otimes \cdots \otimes x_{n-1}\rangle.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value					

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00001	001	—	$\{ \}^{\text{\texttt{\text{@}}}}$	$\text{\textcircled{0}}$
rph8-g15q_F00002	001	—	$\{ \}^{\{1,2,*\}}$	$1,2,*$
rph8-g15q_F00003	001	—	$\{ \}^{\{3,4,\text{\texttt{\text{d}}ag}}\}$	$3,4,\uparrow$
rph8-g15q_F00004	001	—	$\{ \}^{\{5,\text{\texttt{\text{d}}dag}}\}$	$5,\ddagger$
rph8-g15q_F00005		—	$\{ \}^{\text{\texttt{\text{d}}ag}}$	$\uparrow$
rph8-g15q_F00006		—	$\{ \}^{\text{\texttt{\text{d}}dag}}$	$\ddagger$
rph8-g15q_F00007	003	—	x	x
rph8-g15q_F00008	003	—	$\text{\texttt{\texttheta}}$	θ
rph8-g15q_F00009	003	—	$\text{\texttt{\text{r}}angle}$	$\rangle$
rph8-g15q_F00010	003	—	$v=\text{\texttt{\text{left}}}(v_{\text{\texttt{\text{1}}}},\text{\texttt{\text{ldots}}},v_{\text{\texttt{\text{d}}}})\text{\texttt{\text{right}}})$	$v=(v_1,\dots,v_d)$
rph8-g15q_F00011	003	—	$\text{\texttt{\text{mathbb}}}\{R\}^{\text{\texttt{\text{d}}}}$	$\mathbb{R}^d$
rph8-g15q_F00012	003	—	$\text{\texttt{\text{mathcal}}}\{H\}$	$\mathcal{H}$
rph8-g15q_F00013	003	—	$\text{\texttt{\text{mathscr}}}\{S\}(\text{\texttt{\text{mathcal}}}\{H\})$	$\mathscr{S}(\mathcal{H})$
rph8-g15q_F00014	003	—	$\text{\texttt{\text{mathscr}}}\{U\}(\text{\texttt{\text{mathcal}}}\{H\})$	$\mathscr{U}(\mathcal{H})$
rph8-g15q_F00015	003	—	$\text{\texttt{\text{mathcal}}}\{X\}$	$\mathcal{X}$
rph8-g15q_F00016	003	—	X	X
rph8-g15q_F00017	003	—	$\text{\texttt{\text{rho}}}:\text{\texttt{\text{mathcal}}}\{X\}\text{\texttt{\text{rightarrow}}}\text{\texttt{\text{mathscr}}}\{S\}(\text{\texttt{\text{mathcal}}}\{H\})$	$\rho:\mathcal{X}\rightarrow\mathscr{S}(\mathcal{H})$
rph8-g15q_F00018	003	—	$U:\text{\texttt{\text{mathcal}}}\{X\}\text{\texttt{\text{rightarrow}}}\text{\texttt{\text{mathscr}}}\{U\}(\text{\texttt{\text{mathcal}}}\{H\})$	$U:\mathcal{X}\rightarrow\mathscr{U}(\mathcal{H})$
rph8-g15q_F00019	003	—	$\text{\texttt{\text{left}}}\text{\texttt{\text{psi}}}_{\text{\texttt{\text{0}}}}\text{\texttt{\text{right}}}\text{\texttt{\text{rangle}}}\text{\texttt{\text{in}}}\text{\texttt{\text{mathcal}}}\{H\}$	$ \psi_0\rangle\in\mathcal{H}$
rph8-g15q_F00020	003	—	$\text{\texttt{\text{rho}}}:\text{\texttt{\text{mathcal}}}\{X\}\text{\texttt{\text{rightarrow}}}$	$\rho:\mathcal{X}\rightarrow$
rph8-g15q_F00021	004	—	G	G
rph8-g15q_F00022	004	—	$\text{\texttt{\text{alpha}}}:\text{\texttt{\text{mathcal}}}\{X\}\text{\texttt{\text{rightarrow}}}\text{\texttt{\text{operatorname}}}\{\text{\texttt{\text{Aut}}}\}(\text{\texttt{\text{mathcal}}}\{X\})$	$\alpha:G\rightarrow\text{Aut}(\mathcal{X})$
rph8-g15q_F00023	004	—	$\text{\texttt{\text{operatorname}}}\{\text{\texttt{\text{Aut}}}\}(\text{\texttt{\text{mathcal}}}\{X\})$	$\text{Aut}(\mathcal{X})$
rph8-g15q_F00024	004	—	$\text{\texttt{\text{alpha}}}$	α
rph8-g15q_F00025	004	—	$(\text{\texttt{\text{mathcal}}}\{X\},\text{\texttt{\text{alpha}}})$	$(\mathcal{X},\alpha)$
rph8-g15q_F00026	004	—	$g\text{\texttt{\text{in}}}G$	$g\in G$
rph8-g15q_F00027	004	—	$\text{\texttt{\text{alpha}}}_{\text{\texttt{\text{g}}}}:\text{\texttt{\text{mathcal}}}\{X\}\text{\texttt{\text{rightarrow}}}\text{\texttt{\text{mathcal}}}\{X\}$	$\alpha_g:\mathcal{X}\rightarrow\mathcal{X}$
rph8-g15q_F00028	004	—	$x\text{\texttt{\text{in}}}\text{\texttt{\text{mathcal}}}\{X\}$	$x\in\mathcal{X}$
rph8-g15q_F00029	004	—	$\text{\texttt{\text{alpha}}}_{\text{\texttt{\text{g}}}}(x)$	$\alpha_g(x)$
rph8-g15q_F00030	004	—	$\text{\texttt{\text{alpha}}}_{\text{\texttt{\text{g}}}\text{\texttt{\text{h}}}}(x)=\text{\texttt{\text{alpha}}}_{\text{\texttt{\text{g}}}}\text{\texttt{\text{left}}}(\text{\texttt{\text{alpha}}}_{\text{\texttt{\text{h}}}}(x)\text{\texttt{\text{right}}})$	$\alpha_{gh}(x)=\alpha_g(\alpha_h(x))$
rph8-g15q_F00031	004	—	$g,\text{\texttt{\text{h}}}\text{\texttt{\text{in}}}G$	$g,h\in G$
rph8-g15q_F00032	004	—	$\text{\texttt{\text{alpha}}}_{\text{\texttt{\text{1}}}\{G\}}=\text{\texttt{\text{mathrm}}}\{\text{\texttt{\text{id}}}\}_{\text{\texttt{\text{mathcal}}}\{X\}}$	$\alpha_{1_G}=\text{id}_{\mathcal{X}}$
rph8-g15q_F00033	004	—	$1_{\text{\texttt{\text{G}}}}$	1_G
rph8-g15q_F00034	004	—	$V:\text{\texttt{\text{G}}}\text{\texttt{\text{rightarrow}}}$	$V:G\rightarrow$
rph8-g15q_F00035	004	—	$\text{\texttt{\text{operatorname}}}\{\text{\texttt{\text{Aut}}}\}(\text{\texttt{\text{mathcal}}}\{H\})$	$\text{Aut}(\mathcal{H})$
rph8-g15q_F00036	004	—	$\text{\texttt{\text{operatorname}}}\{\text{\texttt{\text{Ad}}}_{\text{\texttt{\text{V}}}\{g\}}\}\text{\texttt{\text{in}}}\text{\texttt{\text{operatorname}}}\{\text{\texttt{\text{Aut}}}\}(\text{\texttt{\text{mathscr}}}\{S\}(\text{\texttt{\text{mathcal}}}\{H\}))$	$\text{Ad}_{V_g}\in\text{Aut}(\mathscr{S}(\mathcal{H}))$
rph8-g15q_F00037	004	—	$\text{\texttt{\text{alpha}}}_{\text{\texttt{\text{(}}1,\text{\texttt{\text{0}}}\text{\texttt{\text{)}}}}}$	$\alpha_{(1,0)}$
rph8-g15q_F00038	004	—	$x_{\text{\texttt{\text{2}}}}=x_{\text{\texttt{\text{1}}}},\text{\texttt{\text{alpha}}}_{\text{\texttt{\text{(}}0,1\text{\texttt{\text{)}}}}}$	$x_2=x_1,\alpha_{(0,1)}$
rph8-g15q_F00039	004	—	$\text{\texttt{\text{alpha}}}_{\text{\texttt{\text{(}}1,1\text{\texttt{\text{)}}}}}$	$\alpha_{(1,1)}$
rph8-g15q_F00040	004	—	$x_{\text{\texttt{\text{2}}}}=-x_{\text{\texttt{\text{1}}}}$	$x_2=-x_1$
rph8-g15q_F00041	004	—	$\text{\texttt{\text{sigma}}}\text{\texttt{\text{in}}}\text{\texttt{\text{mathscr}}}\{S\}(\text{\texttt{\text{mathcal}}}\{H\})$	$\sigma\in\mathscr{S}(\mathcal{H})$
rph8-g15q_F00042	004	—	$\text{\texttt{\text{left}}}(\text{\texttt{\text{mathscr}}}\{S\}(\text{\texttt{\text{mathcal}}}\{H\}),\text{\texttt{\text{operatorname}}}\{\text{\texttt{\text{Ad}}}_{\text{\texttt{\text{V}}}}\}\text{\texttt{\text{right}}})$	$(\mathscr{S}(\mathcal{H}),\text{Ad}_V)$
rph8-g15q_F00043	004	—	$V:\text{\texttt{\text{G}}}\text{\texttt{\text{rightarrow}}}\text{\texttt{\text{mathscr}}}\{U\}(\text{\texttt{\text{mathcal}}}\{H\})$	$V:G\rightarrow\mathscr{U}(\mathcal{H})$
rph8-g15q_F00044	004	—	$\text{\texttt{\text{rho}}}$	ρ
rph8-g15q_F00045	004	—	$\text{\texttt{\text{mathcal}}}\{X\},\text{\texttt{\text{alpha}}}$	$\mathcal{X},\alpha$
rph8-g15q_F00046	004	—	$\text{\texttt{\text{mathcal}}}\{Y\},\text{\texttt{\text{beta}}}$	$\mathcal{Y},\beta$
rph8-g15q_F00047	004	—	$(\text{\texttt{\text{mathcal}}}\{Y\},\text{\texttt{\text{beta}}})$	$(\mathcal{Y},\beta)$
rph8-g15q_F00048	004	—	$f:\text{\texttt{\text{mathcal}}}\{X\}\text{\texttt{\text{rightarrow}}}$	$f:\mathcal{X}\rightarrow$
rph8-g15q_F00049	004	—	$\text{\texttt{\text{mathcal}}}\{Y\}$	$\mathcal{Y}$
rph8-g15q_F00050	005	—	$\text{\texttt{\text{mathcal}}}\{X\}=\text{\texttt{\text{mathbb}}}\{R\}^{\text{\texttt{\text{2}}}}$	$\mathcal{X}=\mathbb{R}^2$
rph8-g15q_F00051	005	—	$y:\text{\texttt{\text{mathcal}}}\{X\}\text{\texttt{\text{rightarrow}}}\text{\texttt{\text{mathbb}}}\{R\}$	$y:\mathcal{X}\rightarrow\mathbb{R}$
rph8-g15q_F00052	005	—	$y(x)>0$	$y(x)>0$
rph8-g15q_F00053	005	—	$y(x)<0$	$y(x)<0$
rph8-g15q_F00054	005	—	$G\text{\texttt{\text{rightarrow}}}\text{\texttt{\text{operatorname}}}\{\text{\texttt{\text{Aut}}}\}\text{\texttt{\text{left}}}(\text{\texttt{\text{mathbb}}}\{R\}^{\text{\texttt{\text{2}}}}\text{\texttt{\text{right}}})$	$G\rightarrow\text{Aut}\left(\mathbb{R}^2\right)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00055	005	—	$\mathcal{H}=\mathbb{C}^2 \otimes \mathbb{C}^2$	$\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$
rph8-g15q_F00056	005	—	$X=\left[\begin{array}{ll}0 & 1 \\ 1 & 0\end{array}\right], \mathbb{K}_2$	$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \mathbb{K}_2$
rph8-g15q_F00057	005	—	2×2	2×2
rph8-g15q_F00058	005	—	$\operatorname{SWAP}(\psi\rangle\otimes \phi\rangle)= \phi\rangle\otimes \psi\rangle$	$\operatorname{SWAP}(\psi\rangle\otimes \phi\rangle)= \phi\rangle\otimes \psi\rangle$
rph8-g15q_F00059	005	—	$ \psi\rangle, \phi\rangle\in\mathbb{C}^2$	$ \psi\rangle, \phi\rangle \in \mathbb{C}^2$
rph8-g15q_F00060	005	—	$V_{\{g\}}\,\rho(x)\,V_{\{g\}}^{\dagger}=\rho\big(\alpha_g(x)\big)$	$V_g\rho(x)V_g^\dagger=\rho\left(\alpha_g(x)\right)$
rph8-g15q_F00061	005	—	$x=\left(x_{\{1\}},\,x_{\{2\}}\right)\in\mathcal{X}$	$x = (x_1, x_2) \in \mathcal{X}$
rph8-g15q_F00062	005	—	Z	Z
rph8-g15q_F00063	005	—	$Z=\left[\begin{array}{cc}1 & 0 \\ 0 & -1\end{array}\right]$	$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
rph8-g15q_F00064	005	—	$0<p<1$	$0 < p < 1$
rph8-g15q_F00065	005	—	p	p
rph8-g15q_F00066	005	—	$ \pm\rangle, \pm\rangle:= \pm\rangle\otimes \pm\rangle$	$ \pm, \pm\rangle := \pm\rangle \otimes \pm\rangle$
rph8-g15q_F00067	005	—	$x=\left(x_{\{1\}},\,x_{\{2\}}\right)$	$x = (x_1, x_2)$
rph8-g15q_F00068	005	—	$\mathcal{L}: \mathbb{R}^2 \rightarrow \mathbb{M}_4$	$\mathcal{L} : \mathbb{R}^2 \rightarrow \mathbb{M}_4$
rph8-g15q_F00069	005	—	$\mathcal{L}(x) \in \mathbb{M}_4$	$\mathcal{L}(x) \in \mathbb{M}_4$
rph8-g15q_F00070	005	—	$\mathbb{M}_4$	$\mathbb{M}_4$
rph8-g15q_F00071	005	—	$-i\,\mathcal{L}(x)\in\mathfrak{su}(4)$	$-i\mathcal{L}(x) \in \mathfrak{su}(4)$
rph8-g15q_F00072	005	—	4×4	4×4
rph8-g15q_F00073	005	—	$\mathcal{L}$	$\mathcal{L}$
rph8-g15q_F00074	005	—	$e_{\{1\}}=(1,0)$	$e_1 = (1, 0)$
rph8-g15q_F00075	005	—	$e_{\{2\}}=(0,1)$	$e_2 = (0, 1)$
rph8-g15q_F00076	005	—	$2\left(4^2-1\right)=30$	$2\left(4^2-1\right)=30$
rph8-g15q_F00077	005	—	$\mathcal{L}(x)$	$\mathcal{L}(x)$
rph8-g15q_F00078	005	—	V_g	V_g
rph8-g15q_F00079	005	—	$g=(1,0)$	$g = (1, 0)$
rph8-g15q_F00080	005	—	$\operatorname{SWAP}_{L_1}\operatorname{SWAP}=L_2$	$\operatorname{SWAP}_{L_1}\operatorname{SWAP} = L_2$
rph8-g15q_F00081	005	—	$\operatorname{SWAP}_{L_2}\operatorname{SWAP}=L_1$	$\operatorname{SWAP}_{L_2}\operatorname{SWAP} = L_1$
rph8-g15q_F00082	005	—	$g=(0,1)$	$g = (0, 1)$
rph8-g15q_F00083	005	—	$(X\otimes X)L_1(X\otimes X)=-L_1$	$(X\otimes X)L_1(X\otimes X)=-L_1$
rph8-g15q_F00084	005	—	$(X\otimes X)L_2(X\otimes$	$(X\otimes X)L_2(X\otimes$
rph8-g15q_F00085	005	—	$X)=-L_2$	$X)=-L_2$
rph8-g15q_F00086	005	—	$\left(L_{\{1\}},\,L_{\{2\}}\right)$	(L_1, L_2)
rph8-g15q_F00087	005	—	$L_{\{1\}}$	L_1
rph8-g15q_F00088	006	—	$\tau_{\mathcal{X}}$	$\tau_{\mathcal{X}}$
rph8-g15q_F00089	006	—	$\left(\mathcal{X},\tau_{\mathcal{X}}\right)$	$(\mathcal{X}, \tau_{\mathcal{X}})$
rph8-g15q_F00090	006	—	$\mathcal{A}$	$\mathcal{A}$
rph8-g15q_F00091	006	—	$\left(\mathcal{X},\tau_{\mathcal{X}},\mathcal{A}\right)$	$(\mathcal{X}, \tau_{\mathcal{X}}, \mathcal{A})$
rph8-g15q_F00092	006	—	$\mathcal{H}=\mathbb{C}^d$	$\mathcal{H} = \mathbb{C}^d$
rph8-g15q_F00093	006	—	$\mathbb{C}\,P^{\wedge d-1}$	$\mathbb{C}\mathbb{P}^{d-1}$
rph8-g15q_F00094	006	—	$\mathcal{X}=\mathbb{R}^d$	$\mathcal{X} = \mathbb{R}^d$
rph8-g15q_F00095	006	—	$x=\left(x_{\{1\}},\,\ldots,\,x_{\{d\}}\right)\in$	$x = (x_1,\ldots,x_d) \in$
rph8-g15q_F00096	006	—	$X_{\{k\}}$	X_k
rph8-g15q_F00097	006	—	k	k
rph8-g15q_F00098	006	—	$\rho:\mathbb{R}^d\rightarrow\mathscr{S}\left(\mathbb{C}^{2^d}\right)$	$\rho:\mathbb{R}^d\rightarrow\mathscr{S}\left(\mathbb{C}^{2^d}\right)$
rph8-g15q_F00099	006	—	Y	Y
rph8-g15q_F00100	006	—	$\vec{n}\cdot\vec{\sigma}$	$\vec{n} \cdot \vec{\sigma}$
rph8-g15q_F00101	006	—	$\vec{n}$	$\vec{n}$
rph8-g15q_F00102	006	—	$\mathbb{R}^3$	$\mathbb{R}^3$
rph8-g15q_F00103	006	—	$\vec{\sigma}=(X,Y,Z)$	$\vec{\sigma} = (X, Y, Z)$
rph8-g15q_F00104	006	—	d	d
rph8-g15q_F00105	007	—	$x\in\mathbb{R}^{2^d}\backslash\{0\}$	$x \in \mathbb{R}^{2^d} \backslash \{0\}$
rph8-g15q_F00106	007	—	$\chi(x)$	$\chi(x)$
rph8-g15q_F00107	007	—	$\mathbb{C}^{2^d}$	$\mathbb{C}^{2^d}$
rph8-g15q_F00108	007	—	$ \Phi(x)\rangle$	$ \Phi(x)\rangle$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00109	007	—	$x=\left(x_{\{0\}},\,x_{\{1\}},\,\ldots,\,x_{\{2^{\{d\}}-1\}}\right)\,\,\text{in}\,\,\mathbb{R}^{\{2^{\{d\}}\}}$	$x=(x_0,x_1,\ldots,x_{2^d-1})\in\mathbb{R}^{2^d}$
rph8-g15q_F00110	007	—	n	n
rph8-g15q_F00111	007	—	$d\,\,0\,\,\mathrm{\sim s}$	$d0\,\,s$
rph8-g15q_F00112	007	—	$d=2$	$d=2$
rph8-g15q_F00113	007	—	$n\,\,\text{in}\,\,\{00,01,10,11\}$	$n\in\{00,01,10,11\}$
rph8-g15q_F00114	007	—	$\ \chi(x)\ $	$\ \chi(x)\ $
rph8-g15q_F00115	007	—	$S^{\{2^{\{d\}}-1\}}$	S^{2^d-1}
rph8-g15q_F00116	007	—	$\mathbb{R}^{\{2^{\{d\}}\}}$	$\mathbb{R}^{2^d}$
rph8-g15q_F00117	007	—	$\log_{\,2}\left(2^{\{d\}}\right)=d$	$\log_2\left(2^d\right)=d$
rph8-g15q_F00118	007	—	$S^{\{2^{\{d\}}\}}$	S^{2^d}
rph8-g15q_F00119	007	—	$\frac{\pi}{4}$	$\frac{\pi}{4}$
rph8-g15q_F00120	007	—	$\frac{\pi}{2}$	$\frac{\pi}{2}$
rph8-g15q_F00121	007	—	$x_{\{0\}}$	x_0
rph8-g15q_F00122	007	—	$S^{\{1\}}$	S^1
rph8-g15q_F00123	007	—	$[-6,6]\,\,\text{cap}\,\,\mathbb{Z}$	$[-6,6]\cap\mathbb{Z}$
rph8-g15q_F00124	008	—	$\mathbb{R}$	$\mathbb{R}$
rph8-g15q_F00125	008	—	$\left(\mathcal{X},\,d_{\mathcal{X}}\right)$	$(\mathcal{X},d_{\mathcal{X}})$
rph8-g15q_F00126	008	—	$d_{\mathcal{X}}:\mathcal{X}\times\mathcal{X}\rightarrow\mathbb{R}$	$d_{\mathcal{X}}:\mathcal{X}\times\mathcal{X}\rightarrow\mathbb{R}$
rph8-g15q_F00127	010	—	$d_{\mathcal{X}}\left(x_1,x_2\right)=0$	$d_{\mathcal{X}}\left(x_1,x_2\right)=0$
rph8-g15q_F00128	010	—	$x_1=x_2$	$x_1=x_2$
rph8-g15q_F00129		—	$d_{\mathcal{X}}\left(x_1,x_2\right)=d_{\mathcal{X}}\left(x_2,x_1\right)$	$d_{\mathcal{X}}\left(x_1,x_2\right)=d_{\mathcal{X}}\left(x_2,x_1\right)$
rph8-g15q_F00130	008	—	$x_1,x_2\,\,\text{in}\,\,\mathcal{X}$	$x_1,x_2\in\mathcal{X}$
rph8-g15q_F00131	008	—	$d_{\mathcal{X}}\left(x_1,x_2\right)\leqslant d_{\mathcal{X}}\left(x_1,x_3\right)+d_{\mathcal{X}}\left(x_2,x_3\right)$	$d_{\mathcal{X}}\left(x_1,x_2\right)\leqslant d_{\mathcal{X}}\left(x_1,x_3\right)+d_{\mathcal{X}}\left(x_2,x_3\right)$
rph8-g15q_F00132	008	—	x_1,x_2,x_3	x_1,x_2,x_3
rph8-g15q_F00133	008	—	$\text{in}\,\,\mathcal{X}$	$\in\mathcal{X}$
rph8-g15q_F00134	008	—	$d_{\mathcal{X}}$	$d_{\mathcal{X}}$
rph8-g15q_F00135	008	—	$X\,\,\text{subse}t\,\,\mathcal{X}$	$X\subseteq\mathcal{X}$
rph8-g15q_F00136	008	—	$\left(X,\,d_X\right)$	(X,d_X)
rph8-g15q_F00137	008	—	$K_{\bullet}\left(X,\,d_X\right)$	$K_{\bullet}(X,d_X)$
rph8-g15q_F00138	008	—	$H_{\bullet}\left(X,\,d_X\right)$	$H_{\bullet}(X,d_X)$
rph8-g15q_F00139	008	—	d_X	d_X
rph8-g15q_F00140	008	—	$d_X\left(x_1,x_2\right)=d_{\mathcal{X}}\left(x_1,x_2\right)$	$d_X\left(x_1,x_2\right)=d_{\mathcal{X}}\left(x_1,x_2\right)$
rph8-g15q_F00141	008	—	$x_1,x_2\in X$	$x_1,x_2\in X$
rph8-g15q_F00142	008	—	$X,\,d_X$	X,d_X
rph8-g15q_F00143	008	—	$H_{\bullet}\left(X,\,d_X\right)[45,86,87]$	$H_{\bullet}(X,d_X)[45,86,87]$
rph8-g15q_F00144	008	—	$\operatorname{dgm}_{\bullet}\left(X,\,d_X\right)$	$\operatorname{dgm}_{\bullet}(X,d_X)$
rph8-g15q_F00145	008	—	S	S
rph8-g15q_F00146	008	—	ϵ	ϵ
rph8-g15q_F00147	008	—	$s\,\,\text{in}\,\,S$	$s\in S$
rph8-g15q_F00148	008	—	$k+1$	$k+1$
rph8-g15q_F00149	008	—	$v_1,v_2,\ldots,v_n$	$v_1,v_2,\ldots,v_n$
rph8-g15q_F00150	008	—	s	s
rph8-g15q_F00151	008	—	$s=\left\{v_{i_1},\ldots,v_{i_{k+1}}\right\}$	$s=\left\{v_{i_1},\ldots,v_{i_{k+1}}\right\}$
rph8-g15q_F00152	008	—	$i_1<\cdots<i_{k+1}$	$i_1<\cdots<i_{k+1}$
rph8-g15q_F00153	008	—	$v_{i_1},\ldots,v_{i_{k+1}}$	$v_{i_1},\ldots,v_{i_{k+1}}$
rph8-g15q_F00154	008	—	$ s\rangle$	$ s\rangle$
rph8-g15q_F00155	008	—	$i_1,\ldots,i_{k+1}$	$i_1,\ldots,i_{k+1}$
rph8-g15q_F00156	008	—	$ 1\rangle$	$ 1\rangle$
rph8-g15q_F00157	008	—	$ 0\rangle$	$ 0\rangle$
rph8-g15q_F00158	008	—	$n=7$	$n=7$
rph8-g15q_F00159	008	—	$k=3$	$k=3$
rph8-g15q_F00160	008	—	$s=\left\{v_2,v_4,v_5\right\}$	$s=\left\{v_2,v_4,v_5\right\}$
rph8-g15q_F00161	008	—	$ s\rangle= 0101100\rangle$	$ s\rangle= 0101100\rangle$
rph8-g15q_F00162	008	—	f	f
rph8-g15q_F00163	008	—	$\left(\mathcal{X},\,d_{\mathcal{X}}\right)\rightarrow\left(\mathcal{Y},\,d_{\mathcal{Y}}\right)$	$(\mathcal{X},d_{\mathcal{X}})\rightarrow(\mathcal{Y},d_{\mathcal{Y}})$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00164	008	—	$\left(\mathrm{cal}\{Y\}, \mathrm{d}_{\mathrm{cal}\{Y\}}\right)$	$(\mathcal{Y}, d_{\mathcal{Y}})$
rph8-g15q_F00165	008	—	$f: \mathrm{cal}\{X\} \rightarrow \mathrm{cal}\{Y\}$	$f: \mathcal{X} \rightarrow \mathcal{Y}$
rph8-g15q_F00166	008	—	$\mathrm{d}_{\mathrm{cal}\{Y\}}\left(f\left(x_{1}\right), f\left(x_{2}\right)\right) \leqslant \mathrm{d}_{\mathrm{cal}\{X\}}\left(x_{1}, x_{2}\right)$	$d_{\mathcal{Y}}\left(f\left(x_1\right), f\left(x_2\right)\right) \leq d_{\mathcal{X}}\left(x_1, x_2\right)$
rph8-g15q_F00167	009	—	$\mathrm{d}_{\mathrm{cal}\{Y\}}\left(f\left(x_{1}\right), f\left(x_{2}\right)\right)=\mathrm{d}_{\mathrm{cal}\{X\}}\left(x_{1}, x_{2}\right)$	$d_{\mathcal{Y}}\left(f\left(x_1\right), f\left(x_2\right)\right)=d_{\mathcal{X}}\left(x_1, x_2\right)$
rph8-g15q_F00168	009	—	$f:\left(X, \mathrm{d}_X\right) \rightarrow\left(\mathrm{cal}\{Y\}, \mathrm{d}_{\mathrm{cal}\{Y\}}\right)$	$f:\left(X, d_X\right) \rightarrow\left(\mathcal{Y}, d_{\mathcal{Y}}\right)$
rph8-g15q_F00169	009	—	$Y:=f(X)$	$Y:=f(X)$
rph8-g15q_F00170	009	—	d_Y	d_Y
rph8-g15q_F00171	009	—	$\mathrm{d}_{\mathrm{cal}\{Y\}}$	$d_{\mathcal{Y}}$
rph8-g15q_F00172	009	—	$K_{\bullet}\left(Y, \mathrm{d}_Y\right)$	$K_{\bullet}\left(Y, d_Y\right)$
rph8-g15q_F00173	009	—	$H_{\bullet}\left(Y, \mathrm{d}_Y\right)$	$H_{\bullet}\left(Y, d_Y\right)$
rph8-g15q_F00174	009	—	$\operatorname{dgm}_{\bullet}\left(Y, \mathrm{d}_Y\right)$	$\operatorname{dgm}_{\bullet}\left(Y, d_Y\right)$
rph8-g15q_F00175	009	—	$\mathrm{cal}\{X\} \rightarrowtail \mathrm{cal}\{Y\}$	$\mathcal{X} \xrightarrow{f} \mathcal{Y}$
rph8-g15q_F00176	009	—	$\mathrm{d}_{\mathrm{cal}\{X\}}: \mathrm{cal}\{X\} \times \mathrm{cal}\{X\} \rightarrow[0, \infty)$	$d_{\mathcal{X}}: \mathcal{X} \times \mathcal{X} \rightarrow[0, \infty)$
rph8-g15q_F00177	009	—	$\left(\mathrm{cal}\{X\}, \mathrm{d}_{\mathrm{cal}\{X\}}\right) \rightarrowtail\left(\mathrm{cal}\{Y\}, \mathrm{d}_{\mathrm{cal}\{Y\}}\right)$	$\left(\mathcal{X}, d_{\mathcal{X}}\right) \xrightarrow{f}\left(\mathcal{Y}, d_{\mathcal{Y}}\right)$
rph8-g15q_F00178	009	—	L_{p}	L_p
rph8-g15q_F00179	010	—	$\left(\mathrm{cal}\{X\}, \mathrm{d}_{\mathrm{cal}\{X\}_{\text {true }}}\right)$	$\left(\mathcal{X}, d_{\mathcal{X}_{\text {true }}}\right)$
rph8-g15q_F00180	010	—	$\mathrm{d}_{\mathrm{cal}\{X\}_{\text {true }}}}$	$d_{\mathcal{X}_{\text {true}}}$
rph8-g15q_F00181	010	—	σ	σ
rph8-g15q_F00182	010	—	ℓ_{1}	ℓ_1
rph8-g15q_F00183	010	—	ℓ_{2}	ℓ_2
rph8-g15q_F00184	010	—	$\rho, \sigma \in \mathscr{S}(\mathrm{cal}\{H\})$	$\rho, \sigma \in \mathscr{S}(\mathcal{H})$
rph8-g15q_F00185	010	—	$\mathrm{d}_{\mathrm{cal}\{H\}}$	$d_{\mathcal{H}}$
rph8-g15q_F00186	010	—	Θ	Θ
rph8-g15q_F00187	010	—	$\rho: \mathrm{cal}\{X\} \times \Theta \rightarrow \mathscr{S}(\mathrm{cal}\{H\})$	$\rho: \mathcal{X} \times \Theta \rightarrow \mathscr{S}(\mathcal{H})$
rph8-g15q_F00188	010	—	$\theta \in \Theta$	$\theta \in \Theta$
rph8-g15q_F00189	010	—	$\mathrm{cal}\{X\}^{\prime}$	$\mathcal{X}'$
rph8-g15q_F00190	010	—	$\mathscr{S}(\mathrm{cal}\{H\})^{\prime}$	$\mathscr{S}(\mathcal{H})'$
rph8-g15q_F00191	010	—	$\rho^{\prime}: \mathrm{cal}\{X\}^{\prime} \rightarrow \mathscr{S}(\mathrm{cal}\{H\})^{\prime}$	$\rho': \mathcal{X}' \rightarrow \mathscr{S}(\mathcal{H})'$
rph8-g15q_F00192	010	—	$\rho^{\prime}$	ρ'
rph8-g15q_F00193	011	—	$\mathrm{cal}\{X\}=[-2,2]$	$\mathcal{X}=[-2,2]$
rph8-g15q_F00194	011	—	$\rho: [-2,2] \rightarrow \mathscr{S}\left(\mathbb{C}^2\right)$	$\rho: [-2,2] \rightarrow \mathscr{S}\left(\mathbb{C}^2\right)$
rph8-g15q_F00195	011	—	$\mathscr{S}\left(\mathbb{C}^2\right)$	$\mathscr{S}\left(\mathbb{C}^2\right)$
rph8-g15q_F00196	011	—	$\theta_1=0.31, \theta_2=1.48, \theta_3=0$	$\theta_1=0.31, \theta_2=1.48, \theta_3=0$
rph8-g15q_F00197	011	—	$\theta_4=0$	$\theta_4=0$
rph8-g15q_F00198	011	—	$\rho: \mathrm{cal}\{X\} \rightarrow \mathscr{S}\left(\mathbb{C}^2\right)$	$\rho: \mathcal{X} \rightarrow \mathscr{S}\left(\mathbb{C}^2\right)$
rph8-g15q_F00199	011	—	$f:$	$f:$
rph8-g15q_F00200	011	—	$\mathrm{cal}\{X\} \rightarrow \mathrm{cal}\{Y\}$	$\mathcal{X} \rightarrow \mathcal{Y}$
rph8-g15q_F00201	011	—	$g: \mathrm{cal}\{Y\} \rightarrow \mathrm{cal}\{Z\}$	$g: \mathcal{Y} \rightarrow \mathcal{Z}$
rph8-g15q_F00202	011	—	$g \circ f: \mathrm{cal}\{X\} \rightarrow \mathrm{cal}\{Z\}$	$g \circ f: \mathcal{X} \rightarrow \mathcal{Z}$
rph8-g15q_F00203	011	—	$f: \mathrm{cal}\{X\} \rightarrow \mathrm{cal}\{Y\}, g: \mathrm{cal}\{Y\} \rightarrow \mathrm{cal}\{Z\}$	$f: \mathcal{X} \rightarrow \mathcal{Y}, g: \mathcal{Y} \rightarrow \mathcal{Z}$
rph8-g15q_F00204	011	—	$h: \mathrm{cal}\{Z\} \rightarrow \mathrm{cal}\{W\}$	$h: \mathcal{Z} \rightarrow \mathcal{W}$
rph8-g15q_F00205	011	—	$h \circ(g \circ f)$	$h \circ(g \circ f)$
rph8-g15q_F00206	011	—	$(h \circ g) \circ f$	$(h \circ g) \circ f$
rph8-g15q_F00207	011	—	$\operatorname{id}_{\mathrm{cal}\{X\}}: \mathrm{cal}\{X\} \rightarrow \mathrm{cal}\{X\}$	$\operatorname{id}_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X}$
rph8-g15q_F00208	011	—	$\operatorname{id}_{\mathrm{cal}\{X\}}(x)=x$	$\operatorname{id}_{\mathcal{X}}(x)=x$
rph8-g15q_F00209	011	—	$\operatorname{id}_{\mathrm{cal}\{Y\}} \circ f=f \circ \operatorname{id}_{\mathrm{cal}\{X\}}=f$	$\operatorname{id}_{\mathcal{Y}} \circ f=f \circ \operatorname{id}_{\mathcal{X}}=f$
rph8-g15q_F00210	011	—	$\mathbf{C}$	$\mathbf{C}$
rph8-g15q_F00211		—	$\mathrm{cal}\{X\}, \mathrm{cal}\{Y\}, \ldots$	$\mathcal{X}, \mathcal{Y}, \ldots$
rph8-g15q_F00212	011	—	$\mathrm{cal}\{X\}, \mathrm{cal}\{Y\}$	$\mathcal{X}, \mathcal{Y}$
rph8-g15q_F00213	011	—	$f: \mathrm{cal}\{X\} \rightarrow \mathrm{cal}\{Y\}, \quad g: \mathrm{cal}\{Y\} \rightarrow \mathrm{cal}\{Z\}, \quad$	$f: \mathcal{X} \rightarrow \mathcal{Y}, \quad g: \mathcal{Y} \rightarrow \mathcal{Z},$
rph8-g15q_F00214	012	—	$f \circ \operatorname{id}_{\mathrm{cal}\{X\}}=$	$f \circ \operatorname{id}_{\mathcal{X}}=$
rph8-g15q_F00215	012	—	$\operatorname{id}_{\mathrm{cal}\{Y\}} \circ f=f$	$\operatorname{id}_{\mathcal{Y}} \circ f=f$
rph8-g15q_F00216	012	—	$\operatorname{id}_{\mathrm{cal}\{X\}}$	$\operatorname{id}_{\mathcal{X}}$
rph8-g15q_F00217	012	—	$\{ \ }^{\text {emb }}$	emb
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00218	012	—	$\{ \}^{\mathrm{nif}}$	nif
rph8-g15q_F00219	012	—	$\mathbf{Met}^{\mathrm{nif}}$	Met ^{nif}
rph8-g15q_F00220	012	—	$\{ \}^{\mathrm{inf}}$	inf
rph8-g15q_F00221	012	—	$\{ \}^{\mathrm{\text{nif}}}$	nif
rph8-g15q_F00222	012	—	$\rho:(\mathcal{X},\,\alpha)\rightarrow\left(\mathscr{S}(\mathcal{H}),\,\operatorname{Ad}_{\mathcal{V}}\right)$	$\rho:(\mathcal{X},\alpha)\rightarrow(\mathscr{S}(\mathcal{H}),\mathrm{Ad}_V)$
rph8-g15q_F00223	012	—	F: G	$F:G$
rph8-g15q_F00224	013	—	$F(\mathcal{X},\,\alpha)=\mathcal{X}$	$F(\mathcal{X},\alpha)=\mathcal{X}$
rph8-g15q_F00225	013	—	$F(f)=f$	$F(f)=f$
rph8-g15q_F00226	013	—	F	F
rph8-g15q_F00227	013	—	g	g
rph8-g15q_F00228	013	—	$x\mapsto x$	$x\mapsto x$
rph8-g15q_F00229	013	—	$F:\mathbf{C}\rightarrow\mathbf{D}$	$F:\mathbf{C}\rightarrow\mathbf{D}$
rph8-g15q_F00230		—	$F(\mathcal{X})$	$F(\mathcal{X})$
rph8-g15q_F00231	014	—	$\mathbf{D}$	D
rph8-g15q_F00232	013	—	$F(f):F(\mathcal{X})\rightarrow F(\mathcal{Y})$	$F(f):F(\mathcal{X})\rightarrow F(\mathcal{Y})$
rph8-g15q_F00233	013	—	$F(g\circ f)=$	$F(g\circ f)=$
rph8-g15q_F00234	013	—	$F(g)\circ F(f)$	$F(g)\circ F(f)$
rph8-g15q_F00235	013	—	$F(\operatorname{id}_{\mathcal{X}})=\operatorname{id}_{F(\mathcal{X})}$	$F(\operatorname{id}_{\mathcal{X}})=\operatorname{id}_{F(\mathcal{X})}$
rph8-g15q_F00236	013	—	$\mathbf{C}(\mathcal{X},\mathcal{Y})$	$\mathbf{C}(\mathcal{X},\mathcal{Y})$
rph8-g15q_F00237	013	—	$\mathbf{C}(\mathcal{X},\mathcal{Y})\rightarrow\mathbf{D}(F(\mathcal{X}),F(\mathcal{Y}))$	$\mathbf{C}(\mathcal{X},\mathcal{Y})\rightarrow\mathbf{D}(F(\mathcal{X}),F(\mathcal{Y}))$
rph8-g15q_F00238	013	—	$f\mapsto F(f)$	$f\mapsto F(f)$
rph8-g15q_F00239	013	—	$\mathbf{C}\rightarrow\mathbf{D}$	$\mathbf{C}\rightarrow\mathbf{D}$
rph8-g15q_F00240	014	—	$F(\mathcal{X}^{\prime})=\mathcal{X}$	$F(\mathcal{X}')=\mathcal{X}$
rph8-g15q_F00241	014	—	$F(\mathscr{S}(\mathcal{H})^{\prime})=\mathscr{S}(\mathcal{H})$	$F(\mathscr{S}(\mathcal{H})')=\mathscr{S}(\mathcal{H})$
rph8-g15q_F00242	014	—	$F:\mathbf{C}\rightarrow$	$F:\mathbf{C}\rightarrow$
rph8-g15q_F00243	014	—	$F(\rho^{\prime})=\rho$	$F(\rho')=\rho$
rph8-g15q_F00244	014	—	$\mathcal{X}^{\prime}:=(\mathcal{X},\alpha)$	$\mathcal{X}':=(\mathcal{X},\alpha)$
rph8-g15q_F00245	014	—	$G\ni g\mapsto V_g\in\mathscr{U}(\mathcal{H})$	$G\ni g\mapsto V_g\in\mathscr{U}(\mathcal{H})$
rph8-g15q_F00246	014	—	$G\ni g\mapsto\operatorname{Ad}_{V_g}$	$G\ni g\mapsto\mathrm{Ad}_{V_g}$
rph8-g15q_F00247	014	—	$\mathscr{S}(\mathcal{H})^{\prime}:=$	$\mathscr{S}(\mathcal{H})':=$
rph8-g15q_F00248	014	—	$\rho^{\prime}:\mathcal{X}^{\prime}\rightarrow$	$\rho':\mathcal{X}'\rightarrow$
rph8-g15q_F00249	014	—	U	U
rph8-g15q_F00250	014	—	$\mathbb{R}^d\rightarrow\mathscr{U}(\mathcal{H})$	$\mathbb{R}^d\rightarrow\mathscr{U}(\mathcal{H})$
rph8-g15q_F00251	014	—	$\mathcal{L}:\mathbb{R}^d\rightarrow\mathcal{B}(\mathcal{H})$	$\mathcal{L}:\mathbb{R}^d\rightarrow\mathcal{B}(\mathcal{H})$
rph8-g15q_F00252	014	—	$x\in\mathbb{R}^d$	$x\in\mathbb{R}^d$
rph8-g15q_F00253	014	—	$\mathcal{B}(\mathcal{H})$	$\mathcal{B}(\mathcal{H})$
rph8-g15q_F00254	014	—	$U:\mathbb{R}^d\rightarrow\mathscr{U}(\mathcal{H})$	$U:\mathbb{R}^d\rightarrow\mathscr{U}(\mathcal{H})$
rph8-g15q_F00255	015	—	$\rho:X\rightarrow\mathscr{S}(\mathcal{H})$	$\rho:X\rightarrow\mathscr{S}(\mathcal{H})$
rph8-g15q_F00256	015	—	ρ_{opt}	ρ_{opt}
rph8-g15q_F00257	016	—	$\mathcal{L}:\mathcal{X}\rightarrow\mathcal{B}(\mathcal{H})$	$\mathcal{L}:\mathcal{X}\rightarrow\mathcal{B}(\mathcal{H})$
rph8-g15q_F00258	016	—	$\rho(x)=U(x)\left \psi_0\right\rangle\left\langle\psi_0\right U(x)^{\dagger}$	$\rho(x)=U(x) \psi_0\rangle\langle\psi_0 U(x)^\dagger$
rph8-g15q_F00259	016	—	$\gamma:\mathbb{R}\rightarrow G$	$\gamma:\mathbb{R}\rightarrow G$
rph8-g15q_F00260	016	—	γ	γ
rph8-g15q_F00261		—	$\gamma(0)=1$	$\gamma(0)=1$
rph8-g15q_F00262		—	$\gamma(s+t)=\gamma(s)\gamma(t)$	$\gamma(s+t)=\gamma(s)\gamma(t)$
rph8-g15q_F00263		—	$s,\,t\in\mathbb{R}$	$s,t\in\mathbb{R}$
rph8-g15q_F00264	016	—	$M\in\mathfrak{g}$	$M\in\mathfrak{g}$
rph8-g15q_F00265	016	—	$\mathfrak{g}$	$\mathfrak{g}$
rph8-g15q_F00266	016	—	$\exp:\mathfrak{g}\rightarrow G$	$\exp:\mathfrak{g}\rightarrow G$
rph8-g15q_F00267	016	—	$t\in\mathbb{R}$	$t\in\mathbb{R}$
rph8-g15q_F00268	016	—	$t=0$	$t=0$
rph8-g15q_F00269	016	—	$M=\left.\frac{d}{dt}\gamma\right _{t=0}$	$M=\left.\frac{d}{dt}\gamma\right _{t=0}$
rph8-g15q_F00270	016	—	$\mathcal{X}=\mathbb{R}$	$\mathcal{X}=\mathbb{R}$
rph8-g15q_F00271	016	—	$x\mapsto e^{-ixL}$	$x\mapsto e^{-ixL}$
rph8-g15q_F00272	016	—	$L\in\mathfrak{su}(2^n)$	$L\in\mathfrak{su}(2^n)$
rph8-g15q_F00273	016	—	$\exp:\mathbb{R}^d\rightarrow\mathbb{R}^d$	$\exp:\mathbb{R}^d\rightarrow\mathbb{R}^d$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00274	016	—	$\exp (x)=x$	$\exp(x) = x$
rph8-g15q_F00275	016	—	$[V, W]:=V\, W-W\, V$	$[V, W] := VW - WV$
rph8-g15q_F00276	016	—	H	H
rph8-g15q_F00277	016	—	$f: G \rightarrow H$	$f : G \rightarrow H$
rph8-g15q_F00278	016	—	$\mathcal{M}: \mathfrak{g} \rightarrow \mathfrak{h}$	$\mathcal{M} : \mathfrak{g} \rightarrow \mathfrak{h}$
rph8-g15q_F00279	016	—	$W \in \mathfrak{g}$	$W \in \mathfrak{g}$
rph8-g15q_F00280	016	—	$\mathcal{M}$	$\mathcal{M}$
rph8-g15q_F00281	016	—	$W_1, W_2 \in \mathfrak{g}$	$W_1, W_2 \in \mathfrak{g}$
rph8-g15q_F00282	017	—	$U: \mathcal{X} \rightarrow \mathscr{U} \left(\mathbb{C}^2 \right)$	$U : \mathcal{X} \rightarrow \mathscr{U} \left(\mathbb{C}^2 \right)$
rph8-g15q_F00283	017	—	$(s, t) \in \mathbb{R}^2$	$(s, t) \in \mathbb{R}^2$
rph8-g15q_F00284	017	—	$\mathcal{M}: \mathbb{R}^2 \rightarrow \mathfrak{su}(2)$	$\mathcal{M} : \mathbb{R}^2 \rightarrow \mathfrak{su}(2)$
rph8-g15q_F00285	017	—	$\mathcal{M}((1,0)) = -i X$	$\mathcal{M}((1,0)) = -iX$
rph8-g15q_F00286	017	—	$\mathcal{M}((0,1)) = -i Y$	$\mathcal{M}((0,1)) = -iY$
rph8-g15q_F00287	017	—	$\mathcal{M}((s, t)) = -i s X - i t Y$	$\mathcal{M}((s, t)) = -isX - itY$
rph8-g15q_F00288	017	—	$\exp : \mathbb{R}^2 \rightarrow \mathbb{R}^2$	$\exp : \mathbb{R}^2 \rightarrow \mathbb{R}^2$
rph8-g15q_F00289	017	—	$\mathbb{R}^2$	$\mathbb{R}^2$
rph8-g15q_F00290	017	—	$\mathbb{R}_\alpha \subset \mathbb{R}^2$	$\mathbb{R}_\alpha \subset \mathbb{R}^2$
rph8-g15q_F00291	017	—	$U _{\mathbb{R}_\alpha} : \mathbb{R}_\alpha \rightarrow \mathscr{U} \left(\mathbb{C}^2 \right)$	$U _{\mathbb{R}_\alpha} : \mathbb{R}_\alpha \rightarrow \mathscr{U} \left(\mathbb{C}^2 \right)$
rph8-g15q_F00292	017	—	$W := \alpha \cdot (X, Y)$	$W := \alpha \cdot (X, Y)$
rph8-g15q_F00293	017	—	X, Y	X, Y
rph8-g15q_F00294	017	—	W	W
rph8-g15q_F00295	017	—	$\mathbb{R} \overset{\alpha}{\rightarrow} \mathbb{R}^2$	$\mathbb{R} \overset{\alpha}{\rightarrow} \mathbb{R}^2$
rph8-g15q_F00296	017	—	$r \in \mathbb{R}$	$r \in \mathbb{R}$
rph8-g15q_F00297	017	—	$\alpha(r) := r \alpha$	$\alpha(r) := r\alpha$
rph8-g15q_F00298	017	—	$\mathscr{U} \left(\mathbb{C}^2 \right)$	$\mathscr{U} \left(\mathbb{C}^2 \right)$
rph8-g15q_F00299	017	—	$U(r \alpha)$	$U(r\alpha)$
rph8-g15q_F00300	017	—	$f: G \rightarrow$	$f : G \rightarrow$
rph8-g15q_F00301	017	—	$\mathbb{R} \overset{\gamma}{\rightarrow} G$	$\mathbb{R} \overset{\gamma}{\rightarrow} G$
rph8-g15q_F00302	017	—	$\mathbb{R} \overset{\gamma}{\rightarrow} G \overset{f}{\rightarrow} H$	$\mathbb{R} \overset{\gamma}{\rightarrow} G \overset{f}{\rightarrow} H$
rph8-g15q_F00303	017	—	$G \overset{f}{\rightarrow} H$	$G \overset{f}{\rightarrow} H$
rph8-g15q_F00304	017	—	$x \in \mathfrak{g}$	$x \in \mathfrak{g}$
rph8-g15q_F00305	017	—	$1_G \in G$	$1_G \in G$
rph8-g15q_F00306	017	—	$\mathcal{M} :=$	$\mathcal{M} :=$
rph8-g15q_F00307	017	—	$Df _{1_G}$	$Df _{1_G}$
rph8-g15q_F00308	017	—	$\gamma(0) = 1_G$	$\gamma(0) = 1_G$
rph8-g15q_F00309	017	—	$\left. \frac{d}{dt} \gamma \right _{t=0} = x \in \mathfrak{g}$	$\left. \frac{d}{dt} \gamma \right _{t=0} = x \in \mathfrak{g}$
rph8-g15q_F00310	017	—	$\left. \frac{d}{dt} (f \circ \gamma) \right _{t=0} = \mathcal{M}(x)$	$\left. \frac{d}{dt} (f \circ \gamma) \right _{t=0} = \mathcal{M}(x)$
rph8-g15q_F00311	017	—	$\left. \frac{d}{dt} \gamma \right _{t=0} = x$	$\left. \frac{d}{dt} \gamma \right _{t=0} = x$
rph8-g15q_F00312	017	—	$f \circ \gamma$	$f \circ \gamma$
rph8-g15q_F00313	017	—	$y \in \mathfrak{h}$	$y \in \mathfrak{h}$
rph8-g15q_F00314	018	—	$G = \mathbb{R}^d$	$G = \mathbb{R}^d$
rph8-g15q_F00315	018	—	$\mathfrak{g} = \mathbb{R}^d$	$\mathfrak{g} = \mathbb{R}^d$
rph8-g15q_F00316	018	—	$\mathbb{R}^d \overset{f}{\rightarrow} G$	$\mathbb{R}^d \overset{f}{\rightarrow} G$
rph8-g15q_F00317	018	—	$\mathcal{M}: \mathbb{R}^d \rightarrow$	$\mathcal{M} : \mathbb{R}^d \rightarrow$
rph8-g15q_F00318	018	—	$f: \mathbb{R}^d \rightarrow G$	$f : \mathbb{R}^d \rightarrow G$
rph8-g15q_F00319	018	—	$\mathbb{R}^d \rightarrow \mathscr{U} \left(\mathbb{C}^n \right)$	$\mathbb{R}^d \rightarrow \mathscr{U} \left(\mathbb{C}^n \right)$
rph8-g15q_F00320	018	—	$\mathcal{L}: \mathbb{R}^d \rightarrow \mathcal{B} \left(\mathbb{C}^n \right)$	$\mathcal{L} : \mathbb{R}^d \rightarrow \mathcal{B} \left(\mathbb{C}^n \right)$
rph8-g15q_F00321	018	—	$\mathbb{R}^d \overset{U}{\rightarrow} \mathscr{U} \left(\mathbb{C}^n \right)$	$\mathbb{R}^d \overset{U}{\rightarrow} \mathscr{U} \left(\mathbb{C}^n \right)$
rph8-g15q_F00322	018	—	$U(x+y) = e^{-i \mathcal{L}(x+y)}$	$U(x+y) = e^{-i\mathcal{L}(x+y)}$
rph8-g15q_F00323	018	—	$U(x)U(y) = e^{-i \mathcal{L}(x)} e^{-i \mathcal{L}(y)}$	$U(x)U(y) = e^{-i\mathcal{L}(x)} e^{-i\mathcal{L}(y)}$
rph8-g15q_F00324	018	—	$X \subseteq \mathbb{R}^2$	$X \subseteq \mathbb{R}^2$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00325	018	—	c: \mathcal{X} \rightarrow \{-1,0,+1\}	$c : \mathcal{X} \rightarrow \{-1, 0, +1\}$
rph8-g15q_F00326	018	—	0 \in	$O \in$
rph8-g15q_F00327	018	—	0	O
rph8-g15q_F00328	018	—	\{-1,+1\}	$\{-1, +1\}$
rph8-g15q_F00329	018	—	\rho(x)	$\rho(x)$
rph8-g15q_F00330	018	—	y	y
rph8-g15q_F00331	018	—	y^{\{-1\}}(\theta) \subseteq \mathcal{X}	$y^{-1}(0) \subseteq \mathcal{X}$
rph8-g15q_F00332	018	—	x_{1}, x_{2} \in \mathbb{R}	$x_1, x_2 \in \mathbb{R}$
rph8-g15q_F00333	018	—	\left \psi_0\right\rangle \in \mathcal{H}=\mathbb{C}^4	$ \psi_0\rangle \in \mathcal{H} = \mathbb{C}^4$
rph8-g15q_F00334	018	—	p=0.99	$p = 0.99$
rph8-g15q_F00335	018	—	\left(x_{1}, x_{2}\right)	(x_1, x_2)
rph8-g15q_F00336	018	—	y\left(x_{1}, x_{2}\right)<0	$y\left(x_1, x_2\right) < 0$
rph8-g15q_F00337	018	—	y\left(x_{1}, x_{2}\right)>0	$y\left(x_1, x_2\right) > 0$
rph8-g15q_F00338	018	—	y\left(x_{1}, x_{2}\right)=	$y\left(x_1, x_2\right) =$
rph8-g15q_F00339	018	—	U: \mathbb{R}^2 \rightarrow \mathscr{U}\left(\mathbb{C}^4\right)	$U : \mathbb{R}^2 \rightarrow \mathscr{U}\left(\mathbb{C}^4\right)$
rph8-g15q_F00340	018	—	U(x)=e^{\{-i \mathcal{L}(x)\}}	$U(x) = e^{-i \mathcal{L}(x)}$
rph8-g15q_F00341	018	—	\mathcal{L}: \mathbb{R}^2 \rightarrow \mathfrak{su}(4)	$\mathcal{L} : \mathbb{R}^2 \rightarrow \mathfrak{su}(4)$
rph8-g15q_F00342	018	—	e_{1}, e_{2}	e_1, e_2
rph8-g15q_F00343	018	—	L_{1}:=\mathcal{L}\left(e_{1}\right)	$L_1 := \mathcal{L}\left(e_1\right)$
rph8-g15q_F00344	018	—	L_{2}:=\mathcal{L}\left(e_{2}\right)	$L_2 := \mathcal{L}\left(e_2\right)$
rph8-g15q_F00345	018	—	a_{11}, a_{22}, a_{33}, a_{44} \in \mathbb{R}, a_{12}, a_{13}, a_{14}, a_{23}, a_{24}, a_{34} \in \mathbb{C}	$a_{11}, a_{22}, a_{33}, a_{44} \in \mathbb{R}, a_{12}, a_{13}, a_{14}, a_{23}, a_{24}, a_{34} \in \mathbb{C}$
rph8-g15q_F00346	018	—	a_{11}+a_{22}+a_{33}+a_{44}=0	$a_{11} + a_{22} + a_{33} + a_{44} = 0$
rph8-g15q_F00347	018	—	L_{2}	L_2
rph8-g15q_F00348	018	—	2(4+2(6)-1)=30	$2(4 + 2(6) - 1) = 30$
rph8-g15q_F00349	018	—	\mathbb{R}^2 \rightarrow \mathscr{U}\left(\mathbb{C}^4\right)	$\mathbb{R}^2 \rightarrow \mathscr{U}\left(\mathbb{C}^4\right)$
rph8-g15q_F00350	018	—	L_{2}=	$L_2 =$
rph8-g15q_F00351	019	—	\left\{X \otimes X, L_{1}\right\}=0	$\{X \otimes X, L_1\} = 0$
rph8-g15q_F00352	019	—	\mathbb{R}^n	$\mathbb{R}^n$
rph8-g15q_F00353	019	—	\mathcal{X}=\mathbb{R}^n	$\mathcal{X} = \mathbb{R}^n$
rph8-g15q_F00354	019	—	\left(\mathcal{Y}=\mathbb{R}^m, d_{\mathcal{Y}}\right)	$(\mathcal{Y} = \mathbb{R}^m, d_{\mathcal{Y}})$
rph8-g15q_F00355	019	—	\mathbb{R}^n \xrightarrow{\varphi} \mathbb{R}^m	$\mathbb{R}^n \xrightarrow{\varphi} \mathbb{R}^m$
rph8-g15q_F00356	019	—	m \times n	$m \times n$
rph8-g15q_F00357	019	—	V	V
rph8-g15q_F00358	019	—	V^{\mathrm{T}}	V^T
rph8-g15q_F00359	019	—	\left(\mathbb{R}^m, A\right)	$(\mathbb{R}^m, A)$
rph8-g15q_F00360	019	—	m \in \mathbb{N}	$m \in \mathbb{N}$
rph8-g15q_F00361	019	—	A	A
rph8-g15q_F00362	019	—	m \times m	$m \times m$
rph8-g15q_F00363	019	—	\mathbb{R}^m, A	$\mathbb{R}^m, A$
rph8-g15q_F00364	019	—	\mathbb{R}^n, B	$\mathbb{R}^n, B$
rph8-g15q_F00365	019	—	n \times m	$n \times m$
rph8-g15q_F00366	019	—	V^{\mathrm{T}} B V=A	$V^T B V = A$
rph8-g15q_F00367	019	—	\mathbb{R}^m	$\mathbb{R}^m$
rph8-g15q_F00368	019	—	\mathbb{R}^m \xrightarrow{V} \mathbb{R}^n	$\mathbb{R}^m \xrightarrow{V} \mathbb{R}^n$
rph8-g15q_F00369	019	—	y \in \mathcal{Y}	$y \in \mathcal{Y}$
rph8-g15q_F00370	019	—	x_{3} \in \mathcal{X}	$x_3 \in \mathcal{X}$
rph8-g15q_F00371	019	—	d_{\mathcal{X}}(x, x)=0	$d_{\mathcal{X}}(x, x) = 0$
rph8-g15q_F00372	019	—	f\left(x_{1}\right)=f\left(x_{2}\right)	$f\left(x_1\right) = f\left(x_2\right)$
rph8-g15q_F00373	019	—	x_{1}, x_{2}	x_1, x_2
rph8-g15q_F00374	019	—	X \subseteq \mathcal{X}	$X \subset \mathcal{X}$
rph8-g15q_F00375	020	—	\{A, B\}	$\{A, B\}$
rph8-g15q_F00376	020	—	B	B
rph8-g15q_F00377	020	—	c: \mathcal{X} \rightarrow \{-1,0,1\}	$c : \mathcal{X} \rightarrow \{-1, 0, 1\}$
rph8-g15q_F00378	020	—	\# A	$\# A$
rph8-g15q_F00379	020	—	\# B	$\# B$
rph8-g15q_F00380	020	—	\mathcal{X} \rightarrow \mathscr{S}(\mathcal{H})	$\mathcal{X} \rightarrow \mathscr{S}(\mathcal{H})$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value				

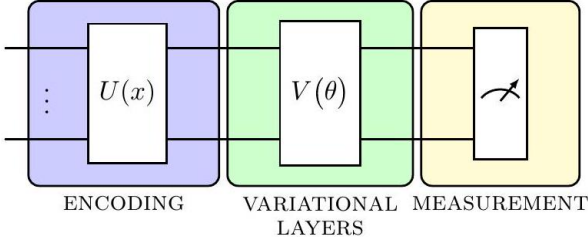
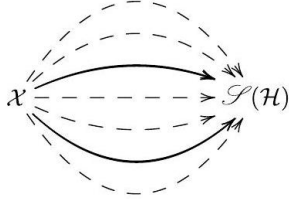
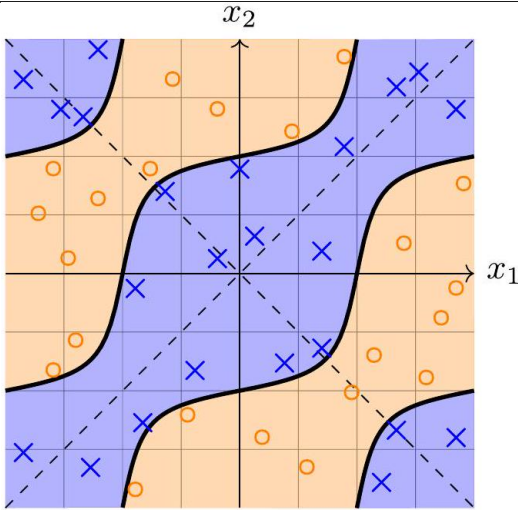
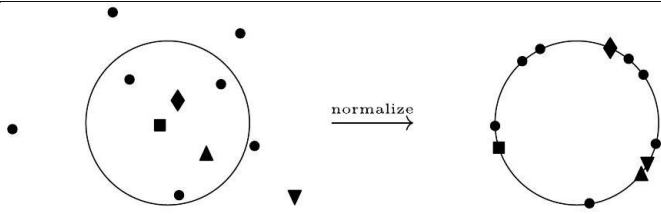
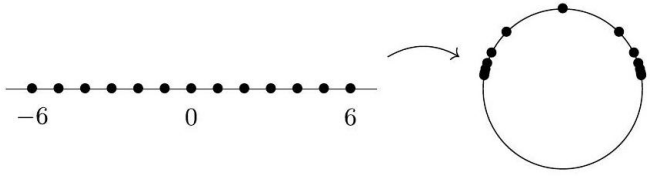
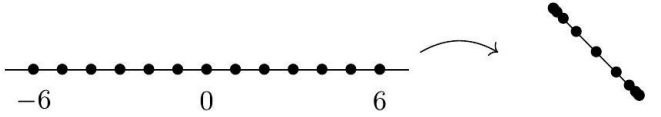
Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00381	020	—	$\rho: \Theta \times \mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$	$\rho: \Theta \times \mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$
rph8-g15q_F00382	020	—	$\Theta = [0, 2 \pi]^4$	$\Theta = [0, 2\pi]^4$
rph8-g15q_F00383	020	—	$\theta = (\theta_1, \theta_2, \theta_3, \theta_4) \in \Theta$	$\theta = (\theta_1, \theta_2, \theta_3, \theta_4) \in \Theta$
rph8-g15q_F00384	020	—	$\rho_{\theta}(x) := \rho(\theta, x)$	$\rho_{\theta}(x) := \rho(\theta, x)$
rph8-g15q_F00385	020	—	$U: \Theta \times \mathcal{X} \rightarrow \mathcal{U}(\mathcal{H})$	$U: \Theta \times \mathcal{X} \rightarrow \mathcal{U}(\mathcal{H})$
rph8-g15q_F00386	020	—	ρ_A	ρ_A
rph8-g15q_F00387	020	—	ρ_B	ρ_B
rph8-g15q_F00388	020	—	$\lambda \in (0, 1]$	$\lambda \in (0, 1]$
rph8-g15q_F00389	020	—	$\rho_A - \rho_B$	$\rho_A - \rho_B$
rph8-g15q_F00390	020	—	$\{-\lambda, +\lambda\}$	$\{-\lambda, +\lambda\}$
rph8-g15q_F00391	020	—	λ_1	λ_1
rph8-g15q_F00392	020	—	λ_2	λ_2
rph8-g15q_F00393	020	—	$\rho_A \neq \rho_B$	$\rho_A \neq \rho_B$
rph8-g15q_F00394	020	—	$\lambda_1 + \lambda_2 = 0$	$\lambda_1 + \lambda_2 = 0$
rph8-g15q_F00395	020	—	$\lambda_2 = -\lambda_1$	$\lambda_2 = -\lambda_1$
rph8-g15q_F00396	020	—	$\lambda = \lambda_1 $	$\lambda = \lambda_1 $
rph8-g15q_F00397	020	—	$A \cup B = X$	$A \cup B = X$
rph8-g15q_F00398	020	—	$O = \rho_A - \rho_B$	$O = \rho_A - \rho_B$
rph8-g15q_F00399	020	—	$\lambda > 0$	$\lambda > 0$
rph8-g15q_F00400	020	—	c	c
rph8-g15q_F00401	020	—	$x \in X$	$x \in X$
rph8-g15q_F00402	020	—	$\operatorname{sgn}:(-\infty, 0) \cup (0, \infty) \rightarrow \{-1, +1\}$	$\operatorname{sgn}:(-\infty, 0) \cup (0, \infty) \rightarrow \{-1, +1\}$
rph8-g15q_F00403	020	—	$\operatorname{sgn}(x)=\frac{x}{ x }$	$\operatorname{sgn}(x)=\frac{x}{ x }$
rph8-g15q_F00404	020	—	$\Pi_+ - \Pi_-$	$\Pi_+ - \Pi_-$
rph8-g15q_F00405	020	—	Π_+	Π_+
rph8-g15q_F00406	020	—	Π_-	Π_-
rph8-g15q_F00407	020	—	$\theta_1, \theta_2, \theta_3$	$\theta_1, \theta_2, \theta_3$
rph8-g15q_F00408	020	—	θ_4	θ_4
rph8-g15q_F00409	020	—	$\rho: \Theta \times \mathcal{X} \rightarrow \mathcal{S}(\mathbb{C}^2)$	$\rho: \Theta \times \mathcal{X} \rightarrow \mathcal{S}(\mathbb{C}^2)$
rph8-g15q_F00410	020	—	$\mathcal{X} \times \mathcal{X}$	$\mathcal{X} \times \mathcal{X}$
rph8-g15q_F00411	021	—	$G: \mathbf{D} \rightarrow \mathbf{E}$	$G: \mathbf{D} \rightarrow \mathbf{E}$
rph8-g15q_F00412	021	—	$G \circ F: \mathbf{C} \rightarrow \mathbf{E}$	$G \circ F: \mathbf{C} \rightarrow \mathbf{E}$
rph8-g15q_F00413	021	—	$G(F(\mathcal{X}))$	$G(F(\mathcal{X}))$
rph8-g15q_F00414	021	—	$G(F(f))$	$G(F(f))$
rph8-g15q_F00415	021	—	F, G	F, G
rph8-g15q_F00416	021	—	η	η
rph8-g15q_F00417	021	—	$\eta: F \Rightarrow G$	$\eta: F \Rightarrow G$
rph8-g15q_F00418	021	—	$\eta_{\mathcal{X}}: F(\mathcal{X}) \rightarrow G(\mathcal{X})$	$\eta_{\mathcal{X}}: F(\mathcal{X}) \rightarrow G(\mathcal{X})$
rph8-g15q_F00419	021	—	$G(f) \circ \eta_{\mathcal{X}} = \eta_{\mathcal{Y}} \circ F(f)$	$G(f) \circ \eta_{\mathcal{X}} = \eta_{\mathcal{Y}} \circ F(f)$
rph8-g15q_F00420	021	—	$\mathcal{C}$	$\mathcal{C}$
rph8-g15q_F00421	021	—	$\mathbb{C}$	$\mathbb{C}$
rph8-g15q_F00422	021	—	$(\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}, 0, +, \cdot)$	$(\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}, 0, +, \cdot)$
rph8-g15q_F00423	021	—	$\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$	$\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$
rph8-g15q_F00424	021	—	v	v
rph8-g15q_F00425	021	—	$\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$	$\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$
rph8-g15q_F00426	021	—	$v: \mathcal{X} \rightarrow \mathbb{C}$	$v: \mathcal{X} \rightarrow \mathbb{C}$
rph8-g15q_F00427	021	—	$v(x) = 0$	$v(x) = 0$
rph8-g15q_F00428	021	—	$v, w \in \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$	$v, w \in \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$
rph8-g15q_F00429	021	—	$v + w$	$v + w$
rph8-g15q_F00430	021	—	$(v + w)(x) :=$	$(v + w)(x) :=$
rph8-g15q_F00431	021	—	$v(x) + w(x)$	$v(x) + w(x)$
rph8-g15q_F00432	021	—	$\lambda \in \mathbb{C}$	$\lambda \in \mathbb{C}$
rph8-g15q_F00433	021	—	$\lambda \cdot v$	$\lambda \cdot v$
rph8-g15q_F00434	021	—	λv	λv
rph8-g15q_F00435	021	—	$\lambda v(x)$	$\lambda v(x)$
rph8-g15q_F00436	021	—	δ_x	δ_x
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00437	021	—	$\delta_{x}\!\left(x^{\mathrm{\scriptscriptstyle\prime}}\right)=\delta_{xx^{\mathrm{\scriptscriptstyle\prime}}}$	$\delta_x(x') = \delta_{xx'}$
rph8-g15q_F00438	021	—	$x^{\mathrm{\scriptscriptstyle\prime}}=x$	$x' = x$
rph8-g15q_F00439	021	—	$x^{\mathrm{\scriptscriptstyle\prime}}\not=$	$x' \neq$
rph8-g15q_F00440	021	—	$ x\rangle$	$ x\rangle$
rph8-g15q_F00441	021	—	$\!\left\langle x^{\mathrm{\scriptscriptstyle\prime}}\mid x\right\rangle=\delta_{xx^{\mathrm{\scriptscriptstyle\prime}}}$	$\langle x' \mid x \rangle = \delta_{xx'}$
rph8-g15q_F00442	021	—	$\mathcal{X}=\mathbb{Z}_2^n=$	$\mathcal{X} = \mathbb{Z}_2^n =$
rph8-g15q_F00443	021	—	$\{0,1\}^n$	$\{0,1\}^n$
rph8-g15q_F00444	021	—	$x=\!\left(x_{\{0\}},\,x_{\{1\}},\,\ldots,\,x_{\{n-1\}}\right)$	$x = (x_0,x_1,\ldots,x_{n-1})$
rph8-g15q_F00445	021	—	$\!\left\{\!0,1,2,\,\ldots,\,2^{\{n\}-1}\right\}$	$\{0,1,2,\ldots,2^n-1\}$
rph8-g15q_F00446	021	—	$\mathbb{C}^{\mathcal{X}}$	$\mathbb{C}^{\mathcal{X}}$
rph8-g15q_F00447	021	—	$2^{\{n\}}$	2^n
rph8-g15q_F00448	021	—	$L_{\{f\}}\colon\mathbb{C}_{\mathrm{\scriptscriptstyle\mathrm{fs}}}^{\mathcal{X}}\rightarrow$	$L_f : \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow$
rph8-g15q_F00449	021	—	$\mathbb{C}_{\mathrm{\scriptscriptstyle\mathrm{fs}}}^{\mathcal{Y}}$	$\mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$
rph8-g15q_F00450	021	—	$ y\rangle$	$ y\rangle$
rph8-g15q_F00451	021	—	$L_{\{f\}}$	L_f
rph8-g15q_F00452	021	—	$ f(x)\rangle$	$ f(x)\rangle$
rph8-g15q_F00453	021	—	$\mathbb{C}^{\mathcal{Y}}$	$\mathbb{C}^{\mathcal{Y}}$
rph8-g15q_F00454	021	—	$L_{\{f\}}\colon\mathbb{C}_{\mathrm{\scriptscriptstyle\mathrm{fs}}}^{\mathcal{X}}\rightarrow\mathbb{C}_{\mathrm{\scriptscriptstyle\mathrm{fs}}}^{\mathcal{Y}}$	$L_f : \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$
rph8-g15q_F00455	021	—	$(V,\,0,+,\,\cdot)$	$(V,0,+,\cdot)$
rph8-g15q_F00456	022	—	$F(\mathcal{X})=$	$F(\mathcal{X}) =$
rph8-g15q_F00457	022	—	$G(F(\mathcal{X}))=\mathbb{C}_{\mathrm{\scriptscriptstyle\mathrm{fs}}}^{\mathcal{X}}$	$G(F(\mathcal{X})) = \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$
rph8-g15q_F00458	022	—	$G\circ F$	$G \circ F$
rph8-g15q_F00459	022	—	$\mathrm{id}_{\mathrm{\scriptscriptstyle\mathrm{Set}}}$	$\mathrm{id}_{\mathrm{Set}}$
rph8-g15q_F00460	022	—	$\eta\colon\mathrm{id}_{\mathrm{\scriptscriptstyle\mathrm{Set}}}\rightarrow$	$\eta : \mathrm{id}_{\mathrm{Set}} \Rightarrow$
rph8-g15q_F00461	022	—	$\eta_{\mathcal{X}}\colon\mathrm{id}_{\mathrm{\scriptscriptstyle\mathrm{Set}}}(\mathcal{X})\rightarrow G(F(\mathcal{X}))$	$\eta_{\mathcal{X}} : \mathrm{id}_{\mathrm{Set}}(\mathcal{X}) \rightarrow G(F(\mathcal{X}))$
rph8-g15q_F00462	022	—	$\eta_{\mathcal{X}}\colon\mathcal{X}\rightarrow\mathbb{C}_{\mathrm{\scriptscriptstyle\mathrm{fs}}}^{\mathcal{X}}$	$\eta_{\mathcal{X}} : \mathcal{X} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$
rph8-g15q_F00463	022	—	$\eta_{\mathcal{X}}(x)= x\rangle\in\mathbb{C}_{\mathrm{\scriptscriptstyle\mathrm{fs}}}^{\mathcal{X}}$	$\eta_{\mathcal{X}}(x) = x\rangle \in \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$
rph8-g15q_F00464	022	—	$\{ x\rangle\}$	$\{ x\rangle\}$
rph8-g15q_F00465	022	—	$\eta_{\mathcal{Y}}(f(x))=L_f(\eta_{\mathcal{X}}(x))$	$\eta_{\mathcal{Y}}(f(x)) = L_f(\eta_{\mathcal{X}}(x))$
rph8-g15q_F00466	022	—	$ f(x)\rangle=L_f x\rangle$	$ f(x)\rangle = L_f x\rangle$
rph8-g15q_F00467	022	—	$\mathbb{B}G$	$\mathbb{B}G$
rph8-g15q_F00468	022	—	$\alpha\colon\mathbb{B}G\rightarrow$	$\alpha : \mathbb{B}G \rightarrow$
rph8-g15q_F00469	022	—	g^{-1}	g^{-1}
rph8-g15q_F00470	022	—	$gg^{-1}=1_G=g^{-1}g$	$gg^{-1} = 1_G = g^{-1}g$
rph8-g15q_F00471	022	—	$\mathrm{id}_{\mathcal{X}}=\alpha_e=\alpha_{gg^{-1}}=\alpha_g\circ\alpha_{g^{-1}}$	$\mathrm{id}_{\mathcal{X}} = \alpha_e = \alpha_{gg^{-1}} = \alpha_g \circ \alpha_{g^{-1}}$
rph8-g15q_F00472	022	—	$\mathrm{id}_{\mathcal{X}}=\alpha_{g^{-1}}\circ\alpha_g$	$\mathrm{id}_{\mathcal{X}} = \alpha_{g^{-1}} \circ \alpha_g$
rph8-g15q_F00473	022	—	α_g	α_g
rph8-g15q_F00474	022	—	$\alpha_{g^{-1}}$	$\alpha_{g^{-1}}$
rph8-g15q_F00475	022	—	α,β	α, β
rph8-g15q_F00476	022	—	$\mathbb{B}G\rightarrow$	$\mathbb{B}G \rightarrow$
rph8-g15q_F00477	022	—	$\mathcal{X}:=\alpha(\bullet)$	$\mathcal{X} := \alpha(\bullet)$
rph8-g15q_F00478	022	—	$\mathcal{Y}:=\beta(\bullet)$	$\mathcal{Y} := \beta(\bullet)$
rph8-g15q_F00479	022	—	$f\colon\alpha\Rightarrow\beta$	$f : \alpha \Rightarrow \beta$
rph8-g15q_F00480	022	—	$\bullet$	$\bullet$
rph8-g15q_F00481	025	—	w^*	w^*
rph8-g15q_F00482	025	—	$x,y,\ldots\in\mathbb{R}$	$x,y,\ldots \in \mathbb{R}$
rph8-g15q_F00483	025	—	$x\rightarrow y$	$x \rightarrow y$
rph8-g15q_F00484	025	—	$x\leqslant y$	$x \leqslant y$
rph8-g15q_F00485	025	—	$\alpha\in\mathbb{R}^2$	$\alpha \in \mathbb{R}^2$
rph8-g15q_F00486	025	—	$\alpha\colon\mathbb{R}\rightarrow\mathbb{R}^2$	$\alpha : \mathbb{R} \rightarrow \mathbb{R}^2$
rph8-g15q_F00487	025	—	$1\in\mathbb{R}$	$1 \in \mathbb{R}$
rph8-g15q_F00488	025	—	$\mathrm{Tr}[\sigma O]$	$\mathrm{Tr}[\sigma O]$
rph8-g15q_F00489	025	—	$\{\sigma\in\mathscr{S}(\mathcal{H})\colon\mathrm{Tr}[\sigma O]=0\}$	$\{\sigma \in \mathscr{S}(\mathcal{H}) : \mathrm{Tr}[\sigma O] = 0\}$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value

Unrecovered image regions — TikZ / table / failed-math candidates

Regions MathPix left as images (no LaTeX). Reconstruct with pdfdrill vision (LLM classify + transcribe to TikZ/tabular/math); verify an LLM result against the real ink with inkdrill.

Identifier	Page	Image	Source
rph8-g15q_DIA_0001	003		tex.zip (local)
rph8-g15q_DIA_0002	004		tex.zip (local)
rph8-g15q_DIA_0003	004		tex.zip (local)
rph8-g15q_DIA_0004	007		tex.zip (local)
rph8-g15q_DIA_0005	007		tex.zip (local)
rph8-g15q_DIA_0006	008		tex.zip (local)
rph8-g15q_DIA_0007	008	$(X, d_X) \mapsto K_\bullet(X, d_X) \mapsto H_\bullet(X, d_X) \mapsto \text{dgm}_\bullet(X, d_X)$ finite metric space filtered simplicial complex persistence vector space persistence diagram	tex.zip (local)

Identifier	Page	Image	Source
rph8-g15q_DIA_0008	011	(a) (b) (c) (d)	tex.zip (local)
rph8-g15q_DIA_0009	013	$G\text{-Set} \xrightarrow{\text{forget action}} \text{Set}$ data domain $(\mathcal{X}, \alpha) \xrightarrow{\text{forget } \alpha} \mathcal{X}$ state space $(\mathcal{S}(\mathcal{H}), \text{Ad}_V) \xrightarrow{\text{forget Ad}_V} \mathcal{S}(\mathcal{H})$ arbitrary G-set $(\mathcal{Y}, \beta) \xrightarrow{\text{forget } \beta} \mathcal{Y}$	tex.zip (local)
rph8-g15q_DIA_0010	014	$\begin{array}{ccc} \mathbf{C} & \mathcal{X}' \dashrightarrow^{\rho'} \mathcal{S}(\mathcal{H})' & \\ F \downarrow & \downarrow & \downarrow \\ \text{Set} & \mathcal{X} \xrightarrow{\rho} \mathcal{S}(\mathcal{H}) & \end{array}$	tex.zip (local)
rph8-g15q_DIA_0011	014	$\begin{array}{ccc} G\text{-Set} & (\mathcal{X}, \alpha) \dashrightarrow^{\rho} (\mathcal{S}(\mathcal{H}), \text{Ad}_V) & \\ F \downarrow & \downarrow & \downarrow \\ \text{Set} & \mathcal{X} \xrightarrow{\rho} \mathcal{S}(\mathcal{H}) & \end{array}$	tex.zip (local)
rph8-g15q_DIA_0012	021	$\begin{array}{ccc} F(\mathcal{X}) & \xrightarrow{F(f)} & F(\mathcal{Y}) \\ \eta_{\mathcal{X}} \downarrow & & \downarrow \eta_{\mathcal{Y}} \\ G(\mathcal{X}) & \xrightarrow{G(f)} & G(\mathcal{Y}) \end{array}$	tex.zip (local)